Impedance Control Design Framework Using Commutative Map Between $SE(3)$ and $\mathfrak{se}(3)$

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***Abstract*—Impedance control is a widely adopted approach that ensures the compliant behavior of robot manipulators as they interact with their environment according to specifically designed dynamics. For tasks involving six degrees of freedom (DoF), it is crucial to appropriately manage the position and orientation of the end-effector by controlling dynamic behavior. However, describing orientational displacement and designing the corresponding rotational impedance can be challenging, especially when we use a minimal representation. The well-known minimal representation for orientation, the Euler angle, suffers from representation singularity. As a remedy, the quaternion or dual quaternion can be an alternative, but with nonminimal representations. This lack of minimal representation, which does not suffer from the representation singularity, often leads to handling the impedance design by directly defining the potential energy function in the matrix Lie group. This article proposes a framework for the six-DoF impedance control design that takes advantage of Lie group theory with minimal representation, known as the exponential coordinate. Since the exponential coordinate can be treated as the Euclidean variable within the injectivity radius, it allows for the formulation of the impedance control more systematically and familiarly. In our framework, a detour strategy is utilized; the impedance is designed in the Lie group $SE(3)$, and the control is designed in the Lie algebra $\mathfrak{se}(3)$, which is isomorphic to the vector space $\mathbb{R}^6$. The group structure of $SE(3)$ can be maintained using the proposed conversion formula between the Lie group and the Lie algebra, called the differential of the exponential map and its time derivative, with a closed-form expression. Experiments with a 6-DoF robot manipulator verified that the proposed impedance control framework effectively reflects the $SE(3)$ group structure and achieves the desired dynamic behavior as the functionality of the impedance control with minimal parameters.**

***Index Terms*—Compliance and impedance control, exponential map, lie algebra, lie group.**

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I. Introduction

ENSURING a compliant behavior is crucial when a robot interacts with its environment. Impedance control is one of the most reliable methods for achieving this, as it involves controlling the dynamic behavior to impose a desired impedance. This approach is widely used in various scenarios that involve frequent contact [1], [2]. In six degrees of freedom (DoF) impedance control, the desired dynamic behavior has both translational and rotational impedance, corresponding to the position and orientation of the robot end-effector, respectively [3]. For typical three-DoF tasks, which focus solely on the relationship between the position and contact force, positional displacement is straightforwardly defined by directly subtracting the actual task position of the end-effector from the equilibrium position. This simplicity has contributed to the widespread adoption of impedance controllers in such scenarios [4], [5], [6].

However, six-DoF tasks, including the orientation of the end-effector, require a suitable representation of orientational displacement. This is challenging because the orientation of the robot end-effector is typically described by a 3×3 rotation matrix, which is the element of the matrix Lie group called *Special Orthogonal Group* $SO(3)$. Since the Lie group is a smooth manifold endowed with a group structure that exhibits special characteristics compared to Euclidean space, a proper description of the orientational displacement is necessary when considering rotational impedance. The usual minimal representation of the orientation with three parameters, such as the Euler angle, has been utilized to handle orientation [7], [8], [9]. However, there are representation singularity issues, such as the gimbal lock singularity [10]. To avoid the representation singularity, the unit quaternion has long been used to represent the orientation since it is a nonminimal representation (four parameters for describing three-DoF rotation). In particular, the imaginary part of the unit quaternion is used for orientational displacement, expressing the relative orientation between the actual and desired frames as a three-dimensional vector, which aligns with three-DoF orientational tasks [11], [12], [13]. However, this approach still encounters a topological obstruction problem, causing the controller to be trapped in unstable equilibrium sets [14], [15].

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Recently, a six-DoF impedance (admittance) controller has been proposed that uses a dual quaternion logarithm map [15], [16]. It resolves the topological obstruction problem in the orientation part by introducing a quaternion logarithmic variable, achieved using quaternion algebra. Still, the dual quaternion is also a nonminimal representation (eight parameters) for six-DoF rigid body motion. It is due to the lack of minimal representations that do not suffer from the representation singularity.

Recall that the motion of the robot end-effector is described by a 4×4 homogeneous transformation matrix, which is an element of *Special Euclidean Group* $SE(3)$. However, since a Lie group is a differentiable manifold, conventional operations such as differentiation and integration, which are valid in Euclidean spaces, are no longer applicable [17], [18]. As a result, designing the impedance control within the framework of a Lie group requires special tools that handle quantities in the Lie group.

A straightforward consideration when designing the impedance controller in the context of the Lie group $SE(3)$ is to derive the potential energy function stored in the spring directly in the Lie group. In [19], [20], [21], and [22], the impedance control is formulated by defining the potential energy function with a spatial spring design [23] in $SE(3)$. A similar perspective is also adopted in [24], [25], and [26], leading to the proposal of geometric impedance control. This is achieved by defining the potential energy function through the Frobenius norm of the homogeneous transformation matrix, an element of the Lie group. Both studies share the idea that the potential energy function is designed directly in the Lie group $SE(3)$. However, such a group-specific potential energy formulation often causes limitations, especially in the orientation part. Since the trace function with the rotation matrix defines the rotational potential energy in $SO(3)$, the rotation angle is not linear to the rotational spring moment [27]. In fact, the rotational spring moment (control input generated by the orientational displacement) vanishes when the rotation angle becomes a half turn (π rad). Therefore, the impedance controller does not converge even with a nonzero orientational displacement in this case. Rather, recent controller designs using the Lie algebra quantities (logarithmic variables) have been proposed [27], [28], [29]. They are consistent formulations with more favorable convergence properties in Euclidean space due to the vector space isomorphism of the Lie algebra. However, they still have difficulty reshaping the robot's inertia since a simple proportional derivative (PD)-based impedance control law is utilized without force feedback. Even though the simple PD control is widely used as an implicit impedance control due to its easy implementation and no requirement for a force/torque (F/T) sensor, it is well-known that it cannot modify the robot's inertia [9], [25]. Inherently, it uses the actual robot inertia as the desired inertia, so it differs from the original impedance control motivation that renders a new desired dynamics.

In this article, the six-DoF impedance controller is derived from the potential energy function design with the six-dimensional exponential coordinate, which is an element of the Lie algebra. Building on our previous work [18], [30], we have developed a commutative mapping between the body twist and the time derivative of the exponential coordinate, providing a closed-form formula for the differential of an exponential map and its time derivative, illustrated in Fig. 1. Since the Lie algebra is isomorphic to a vector space, handling time derivative operations of the exponential coordinates up to the acceleration level is feasible. This alignment allows the impedance control formulation to be consistent with the operations in vector space by rendering the mass, spring, and damper system while achieving a minimal representation. In summary, the proposed work makes the following contributions.

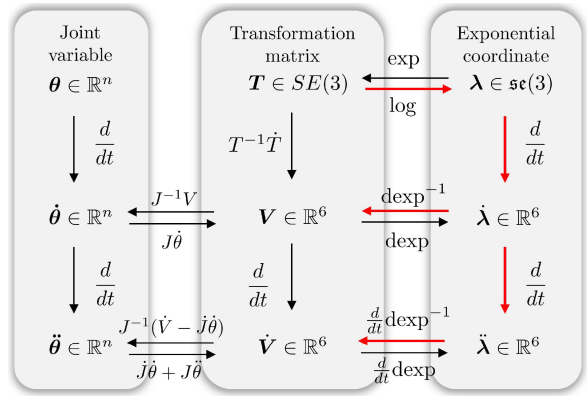

Fig. 1. Commutative map describes the Jacobian relations among three spaces in robot control. Note that the derivation of the body twist requires complex calculation $T^{-1}\dot{T}$ using the homogeneous transformation matrix, which is an element of the Lie group $SE(3)$. Instead, we can take the detour strategy, which is utilized by following the red arrows in this article; that is, utilization of the exponential coordinate, which is an element of the Lie algebra $\mathfrak{se}(3)$ that is isomorphic to the vector space $\mathbb{R}^6$. Then, all operations can be done by $\frac{d}{dt}$ in the vector space $\mathbb{R}^6$. To enable the detour strategy, we need a mapping between the body twist and the time derivative of the exponential coordinate. They are referred to as *differential of exponential map* and described in Section II.

1) A six-DoF impedance control with a minimal representation is proposed by exploiting the exponential coordinate with Lie group theory.
2) An impedance relation is derived from a potential energy function, computed by a six-dimensional exponential coordinate that is an element of the Lie algebra $\mathfrak{se}(3)$.
3) A stability analysis is performed for the impedance relation expressed with the exponential coordinate.

II. Mathematical Background

A. Lie Group and Lie Algebra

A *Lie group* is a group G, which is a differentiable manifold, and the product and inversion are differentiable mappings ($\mu : G \times G \to G$ and $i : G \to G$). Let us consider *matrix Lie groups* as subgroups of $GL(n)$ with the usual matrix product and inversion as the group operation.[1] It is well known that the groups of rotation matrices and homogeneous transformation matrices in the robot manipulator are Lie groups defined by

$$SO(3) = \{R \in \mathbb{R}^{3\times 3} \,|\, R^T R = I, \det(R) = 1\}$$

$$SE(3) = \{T = \begin{bmatrix} R & r \\ 0_{1\times 3} & 1 \end{bmatrix} \in \mathbb{R}^{4\times 4} \,|\, R \in SO(3),\ r \in \mathbb{R}^3\}$$
$$= SO(3) \rtimes \mathbb{R}^3$$

[1] $GL(n)$ represents a general linear group composed of nonsingular $n \times n$ matrices.

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where $0_{1\times 3}$ implies a three-dimensional zero-row vector. Note that $SE(3)$ is the semidirect product group of $\mathbb{R}^3$ and $SO(3)$ represented by the operation $\rtimes$, which has the property that rotational motion affects translation. The homogeneous transformation matrix is sometimes denoted by $T = (R, r) \in SE(3)$ for simplicity.

We can define the tangent space at the identity I in the Lie group. If the Lie bracket can be defined in the tangent space, it is called the Lie algebra, denoted by $\mathfrak{g} = T_I G$. In particular, the Lie algebra of $SO(3)$, which is the group of 3×3 skew-symmetric matrices, denoted by $\mathfrak{so}(3)$, is isomorphic to $\mathbb{R}^3$. In addition, the Lie algebra of $SE(3)$, which is the group of 4×4 matrix representation of twist, denoted by $\mathfrak{se}(3)$, is isomorphic to $\mathbb{R}^6$. For $\omega = (\omega_1, \omega_2, \omega_3)^T \in \mathbb{R}^3$

$$\lceil \omega \rceil = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \in \mathfrak{so}(3). \tag{1}$$

Similarly, for $V = (v^T, \omega^T)^T \in \mathbb{R}^6$

$$\lceil V \rceil = \begin{bmatrix} \lceil \omega \rceil & v \\ 0_{1\times 3} & 0 \end{bmatrix} \in \mathfrak{se}(3) \tag{2}$$

where $\lceil \cdot \rceil$ is called a ceiling form[2] in this article. The Lie bracket for $\mathfrak{se}(3)$, denoted by $[\cdot, \cdot] : \mathfrak{se}(3) \times \mathfrak{se}(3) \to \mathfrak{se}(3)$, is defined by $[\lceil V_1 \rceil, \lceil V_2 \rceil] = \lceil V_1 \rceil \lceil V_2 \rceil - \lceil V_2 \rceil \lceil V_1 \rceil$. To define the adjoint operator,[3] which is equivalent to the Lie bracket operation in $\mathbb{R}^6$, it is calculated as

$$\mathrm{ad}_V = \begin{bmatrix} \lceil \omega \rceil & \lceil v \rceil \\ 0_3 & \lceil \omega \rceil \end{bmatrix} \tag{3}$$

for $V = (v^T, \omega^T)^T \in \mathbb{R}^6$ and 0_3 implies a three-dimensional zero matrix. Then, the Lie bracket can be represented by

$$[\lceil V_1 \rceil, \lceil V_2 \rceil] = \lceil \mathrm{ad}_{V_1} V_2 \rceil. \tag{4}$$

Note that the adjoint operator can also be derived when considering the time derivative of the adjoint transformation. The adjoint transformation plays the role of a coordinate transformation between two frames. It is denoted by Ad_T under the homogeneous transformation matrix $T = (R, r)$, and calculated as

$$\mathrm{Ad}_T = \begin{bmatrix} R & \lceil r \rceil R \\ 0_3 & R \end{bmatrix}. \tag{5}$$

The adjoint operator[4] is also calculated as

$$\mathrm{ad}_V = \mathrm{Ad}_T^{-1} \frac{d}{dt} (\mathrm{Ad}_T) \tag{6}$$

where $\lceil V \rceil = T^{-1} \dot{T}$.

[2]Here, we overload the ceil operator. When applied to a vector $\omega \in \mathbb{R}^3$, it yields 3×3 skew-symmetric matrix of $\mathfrak{so}(3)$. When applied to a twist $V \in \mathbb{R}^6$, this yields 4×4 matrix representation of $\mathfrak{se}(3)$.

[3]For $\mathfrak{so}(3)$, the adjoint operation is equivalent to the ceiling form, i.e., $\mathrm{ad}_\xi = \lceil \xi \rceil$ for $\xi \in \mathbb{R}^3$.

[4]The adjoint operation in (6) is used when the twist is computed through the left translation. If the twist is obtained from the right translation, i.e., $\lceil V \rceil = \dot{T} T^{-1}$, the adjoint operator should be calculated as $\mathrm{ad}_V = \frac{d}{dt}(\mathrm{Ad}_T)\mathrm{Ad}_T^{-1}$. Consistency should be kept. For example, (62) is computed by using the adjoint operation from the right translation.

B. Exponential Map on Lie Group

The exponential map is defined from the Lie algebra $\mathfrak{g}$ to the Lie group G, denoted by $\exp(g) \in G$ for $g \in \mathfrak{g}$. Since we are dealing with a matrix Lie group, the exponential map is aligned with the matrix exponential. It is represented by the ordinary infinite series expansion $\mathrm{expm}(A) = \sum_{k=0}^{\infty} \frac{A^k}{k!}$. For the three-dimensional vector $\xi \in \mathbb{R}^3$, we have a ceiling form $\lceil \xi \rceil \in \mathfrak{so}(3)$. Similarly, for the six-dimensional vector $\lambda = (\eta^T, \xi^T)^T \in \mathbb{R}^6$, we have a ceiling form $\lceil \lambda \rceil \in \mathfrak{se}(3)$. In other words, the exponential maps are defined as

$$\exp(\xi) \triangleq \mathrm{expm}(\lceil \xi \rceil) \in SO(3)$$
$$\exp(\lambda) \triangleq \mathrm{expm}(\lceil \lambda \rceil) \in SE(3).$$

In the case of $SO(3)$, the exponential map has a well-known closed-form expression called Rodrigues' formula [3]

$$\exp(\xi) = I_3 + \frac{\sin(\|\xi\|)}{\|\xi\|} \lceil \xi \rceil + \frac{\sin^2(\|\xi\|/2)}{\|\xi\|^2/2} \lceil \xi \rceil^2 \tag{7}$$

where $\xi = \|\xi\| \hat{\xi}$ and $\hat{\xi}$ is a unit vector of ξ, in which ξ is the equivalent axis for rotation. The above formula can be rewritten [18], [30] as

$$\exp(\xi) = I_3 + \alpha \lceil \xi \rceil + \frac{\beta}{2} \lceil \xi \rceil^2 \tag{8}$$

where

$$\alpha \triangleq sc \tag{9a}$$
$$\beta \triangleq s^2 \tag{9b}$$

with $s \triangleq \frac{\sin(\|\xi\|/2)}{\|\xi\|/2}$ and $c \triangleq \cos(\|\xi\|/2)$. It is observed that $\exp(\xi) = I_3$ when $\xi = 0_{3\times 1}$, where I_3 implies a three-dimensional identity matrix. In the case of $SE(3)$, the exponential map also has the closed-form expression [18] as

$$\exp(\lambda) = \begin{bmatrix} \exp(\xi) & \frac{1}{\|\xi\|^2}\left((I_3 - \exp(\xi))(\xi \times \eta) + \xi^T \xi \eta\right) \\ 0_{1\times 3} & 1 \end{bmatrix}$$
$$\triangleq \begin{bmatrix} \exp(\xi) & \mathrm{dexp}_\xi \eta \\ 0_{1\times 3} & 1 \end{bmatrix} \tag{10}$$

where dexp_ξ is 3×3 matrix as a function of ξ, which is explained later in Section II-C.

C. Exponential Coordinate and Its Differentials

When either $R \in SO(3)$ or $T \in SE(3)$ is given, the inverse image[5] of the exponential map can be considered as either $\xi \in \mathbb{R}^3$ or $\lambda \in \mathbb{R}^6$. These are referred to as the exponential coordinates of R and T, respectively, and are elements of the Lie algebra. For a given rotation matrix R, its corresponding exponential coordinate is calculated as

$$\xi = \log(R) \in \mathbb{R}^3. \tag{11}$$

[5]The inverse map of $\exp(\cdot)$ is referred as $\log(\cdot)$.

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For a homogeneous transformation matrix $T = (R, r)$, the corresponding exponential coordinate is calculated as

$$\lambda = \log(T) = \begin{bmatrix} \eta \\ \xi \end{bmatrix} \in \mathbb{R}^6 \tag{12}$$

where ξ is obtained in (11). Here, the linear component $\eta \in \mathbb{R}^3$ in λ can be computed by the equation $\eta = \mathrm{dexp}_{\xi}^{-1} r$ from (10). Note that the exponential coordinate ξ can be mapped within the region of $\|\xi\| < \pi$, which is called the injectivity radius (or often referred to as the open ball with radius π).

The left translation property is utilized to compute the body angular velocity or the body twist. Given $T(t)$, the associated body twist $V_b(t) = (v_b^T(t), \omega_b^T(t))^T \in \mathbb{R}^6$ is defined from $\lceil V_b(t) \rceil = T^{-1}(t)\dot{T}(t) \in \mathfrak{se}(3)$, where $v_b(t) = R^T(t)\dot{r}(t)$ and $\lceil \omega_b(t) \rceil = R^T(t)\dot{R}(t) \in \mathfrak{so}(3)$. Since $\omega_b(t)$ or $V_b(t)$ is obtained from the time derivative of the Lie group element $R(t)$ or $T(t)$, it might have a connection to the time derivative of the corresponding element in the Lie algebra, i.e., the time derivative of the exponential coordinate $\dot{\xi}(t)$ or $\dot{\lambda}(t)$. It is revealed that there is a mapping called the differential of the exponential map. The differential of the exponential map is defined as the tangent of the exponential map [17], that is, as a function $\mathrm{dexp} : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ such that

$$\frac{d}{dt}\mathrm{expm}(A(t)) = \mathrm{dexp}_{A(t)}\left(\dot{A}(t)\right)\mathrm{expm}(A(t)) \tag{13a}$$

$$= \mathrm{expm}(A(t))\mathrm{dexp}_{-A(t)}\left(\dot{A}(t)\right) \tag{13b}$$

where $A(t)$ is a matrix. If we compute dexp_A as an analytic function using the matrix transformation ad_A [17], [31], i.e.,

$$\mathrm{dexp}_A = \frac{\mathrm{expm}(\mathrm{ad}_A) - I}{\mathrm{ad}_A} = I + \frac{1}{2!}\mathrm{ad}_A + \frac{1}{3!}\mathrm{ad}_A^2 + \cdots \tag{14}$$

it can be expressed as an infinite series by

$$\mathrm{dexp}_A C = \sum_{j=0}^{\infty} \frac{1}{(j+1)!}\mathrm{ad}_A^j C \tag{15}$$

$$\mathrm{dexp}_A^{-1} C = \sum_{j=0}^{\infty} \frac{B_j}{j!}\mathrm{ad}_A^j C \tag{16}$$

where B_j is jth Bernoulli number for $j = 0, 1, 2, \ldots$.

The differential of exponential acts as a mapping between the angular velocity or the twist and time derivative of the exponential coordinates on $SO(3)$ or $SE(3)$. Utilizing (4), we can formally induce the differential of the exponential map such that

$$\frac{d}{dt}\exp(a(t)) = \lceil \mathrm{dexp}_{a(t)}\dot{a}(t) \rceil \exp(a(t)) \tag{17a}$$

$$= \exp(a(t))\lceil \mathrm{dexp}_{-a(t)}\dot{a}(t) \rceil \tag{17b}$$

where $a(t)$ is the associated exponential coordinates of the Lie groups. For given $R(t) = \exp(\xi(t)) \in SO(3)$, plug $\lceil \omega_b(t) \rceil = R^T(t)\dot{R}(t)$ into (17b)

$$\lceil \omega_b(t) \rceil = R^T(t)\dot{R}(t) = \exp(\xi(t))^{-1}\frac{d}{dt}\exp(\xi(t))$$
$$= \lceil \mathrm{dexp}_{-\xi(t)}\dot{\xi}(t) \rceil.$$

Then, we have the relation between body angular velocity and time derivative of three-dimensional exponential coordinates as follows:

$$\omega_b(t) = \mathrm{dexp}_{-\xi(t)}\dot{\xi}(t) \tag{18}$$

$$\dot{\xi}(t) = \mathrm{dexp}_{-\xi(t)}^{-1}\omega_b(t). \tag{19}$$

Similarly, for given $T(t) = \exp(\lambda(t)) \in SE(3)$, we have the relation between body twist and time derivative of six-dimensional exponential coordinates as follows:

$$V_b(t) = \mathrm{dexp}_{-\lambda(t)}\dot{\lambda}(t) \tag{20}$$

$$\dot{\lambda}(t) = \mathrm{dexp}_{-\lambda(t)}^{-1}V_b(t). \tag{21}$$

Note that the differential of the exponential map should be consistent with the description of the coordinate frame of the angular velocity or the twist. Strictly speaking, the body angular velocity $\omega_b(t) \in \mathbb{R}^3$ and the body twist $V_b(t) \in \mathbb{R}^6$, which are obtained using the left translation relationships, $\dot{R}(t) = R(t)\lceil \omega_b(t) \rceil$ and $\dot{T}(t) = T(t)\lceil V_b(t) \rceil$, are linked to the time derivative of the exponential coordinate through left-trivialized definition as in (17b), that is, $\mathrm{dexp}_{-\xi(t)}$ and $\mathrm{dexp}_{-\lambda(t)}$ as in (18) and (20). In contrast, consider the spatial angular velocity $\omega_s(t) \in \mathbb{R}^3$ and the spatial twist $V_s(t) \in \mathbb{R}^6$. Then, the right-trivialized definition of dexp as in (17a), is used to link the time derivative of the exponential coordinates, that is

$$\omega_s(t) = \mathrm{dexp}_{\xi(t)}\dot{\xi}(t) \tag{22}$$

$$V_s(t) = \mathrm{dexp}_{\lambda(t)}\dot{\lambda}(t). \tag{23}$$

Here, the spatial angular velocity and the twist are obtained based on the right translation relationships, $\dot{R}(t) = \lceil \omega_s(t) \rceil R(t)$ and $\dot{T}(t) = \lceil V_s(t) \rceil T(t)$. For this reason, the following relations are satisfied:

$$\mathrm{dexp}_{\xi(t)} = R(t)\mathrm{dexp}_{-\xi(t)} \tag{24}$$

$$\mathrm{dexp}_{\lambda(t)} = \mathrm{Ad}_{T(t)}\mathrm{dexp}_{-\lambda(t)} \tag{25}$$

where $\mathrm{Ad}_{T(t)}$ is the adjoint transformation in (5). In summary, if we consider the homogeneous transformation matrix $T = {}^sT_b = \exp(\lambda)$, the body twist V_b is associated with $\mathrm{dexp}_{-\lambda}$ while the spatial twist V_s is associated with dexp_{λ} to the time derivative of the exponential coordinate $\dot{\lambda}$.

The closed-form of the differential of exponential dexp_{λ} and its time derivative were introduced in [18] and [30] without proof. For $SO(3)$, the differential of the exponential map $\mathrm{dexp}_{-\xi}$ was also reported in [32], and [33] as a form of

$$A(r) = I_3 - \frac{1 - \cos\|r\|}{\|r\|^2}\lceil r \rceil + \frac{\|r\| - \sin\|r\|}{\|r\|^3}\lceil r \rceil^2.$$

Here, r is the same expression as the exponential coordinate ξ, and it is not difficult to show that $\omega_b = A(r)\dot{r}$ is the equivalent expression to (18). Recently, the closed-form of dexp_{λ} and its derivative were introduced in [34] and [35] in the area of computational multibody dynamics. The closed-form formulas for the differential of the exponential map on $SO(3)$ and $SE(3)$ in [18] and [30] are summarized below with proofs in Appendix A.

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Lemma 1 (Differential of exponential map on $SO(3)$): For given exponential map, $R = \exp(\xi)$, with $R \in SO(3)$ and the exponential coordinate $\xi \in \mathbb{R}^3$, the differential of exponential map on $SO(3)$ is given by

$$\mathrm{dexp}_\xi = I_3 + \frac{\beta}{2}\lceil\xi\rceil + \frac{(1-\alpha)}{\|\xi\|^2}\lceil\xi\rceil^2 \tag{26}$$

$$\mathrm{dexp}_\xi^{-1} = I_3 - \frac{1}{2}\lceil\xi\rceil + \frac{(1-\gamma)}{\|\xi\|^2}\lceil\xi\rceil^2 \tag{27}$$

where $\gamma = \frac{\alpha}{\beta}$ from (9a) and (9b). It is noticed that $\mathrm{dexp}_\xi = \mathrm{dexp}_\xi^{-1} = I_3$ when $\xi = 0_{3\times1}$. Furthermore, (26) is defined everywhere (i.e., it converges to the infinite series regardless of the value of ξ), but loses the rank when $\|\xi\| = 2k\pi$ for $k = 1, 2, \cdots$. Correspondingly, (27) is defined only when $\|\xi\| < 2\pi$ (strictly speaking, it converges to the infinite series except $\|\xi\| = 2k\pi$ for $k = 1, 2, \ldots$).

Lemma 2: (Differential of exponential map on $SE(3)$) For given exponential map, $T = \exp(\lambda)$, with $T \in SE(3)$ and exponential coordinates $\lambda = (\eta^T, \xi^T)^T \in \mathbb{R}^6$, the differential of exponential map on $SE(3)$ is given by

$$\mathrm{dexp}_\lambda = \begin{bmatrix} \mathrm{dexp}_\xi & C_\xi(\eta) \\ 0_3 & \mathrm{dexp}_\xi \end{bmatrix} \tag{28}$$

$$\mathrm{dexp}_\lambda^{-1} = \begin{bmatrix} \mathrm{dexp}_\xi^{-1} & D_\xi(\eta) \\ 0_3 & \mathrm{dexp}_\xi^{-1} \end{bmatrix} \tag{29}$$

where

$$\begin{aligned} C_\xi(\eta) = {} & \frac{\beta}{2}\lceil\eta\rceil + \frac{1-\alpha}{\|\xi\|^2}\lceil\eta,\xi\rceil + \frac{\alpha-\beta}{\|\xi\|^2}\xi^T\eta\lceil\xi\rceil \\ & - \frac{1}{\|\xi\|^2}\left(\frac{3(1-\alpha)}{\|\xi\|^2} - \frac{\beta}{2}\right)\xi^T\eta\lceil\xi\rceil^2 \end{aligned} \tag{30}$$

$$\begin{aligned} D_\xi(\eta) = {} & -\frac{1}{2}\lceil\eta\rceil + \frac{1-\gamma}{\|\xi\|^2}\lceil\eta,\xi\rceil \\ & + \frac{1}{\|\xi\|^2}\frac{(1/\beta+\gamma-2)}{\|\xi\|^2}\xi^T\eta\lceil\xi\rceil^2 \end{aligned} \tag{31}$$

in which the notation $\lceil a, b\rceil \triangleq \lceil a\rceil\lceil b\rceil + \lceil b\rceil\lceil a\rceil$ is used for simplicity. If $\xi = 0_{3\times1}$, $C_0(y) = \frac{1}{2}\lceil y\rceil$ and $D_0(y) = -\frac{1}{2}\lceil y\rceil$.

D. Time-Derivative of Differential of Exponential Map

Let us take time derivatives of (18) and (20); the body angular acceleration and the body twist derivative are obtained as

$$\dot{\omega}_b(t) = \mathrm{dexp}_{-\xi(t)}\ddot{\xi}(t) + \left(\frac{d}{dt}\mathrm{dexp}_{-\xi(t)}\right)\dot{\xi}(t) \tag{32}$$

$$\dot{V}_b(t) = \mathrm{dexp}_{-\lambda(t)}\ddot{\lambda}(t) + \left(\frac{d}{dt}\mathrm{dexp}_{-\lambda(t)}\right)\dot{\lambda}(t). \tag{33}$$

To calculate the above equations, the time derivative of the differential of the exponential map should be computed. The subsequent Lemmas will summarize the closed-form formulas of the time derivative of the differential of the exponential map on $SO(3)$ and $SE(3)$.

Lemma 3 (Time-derivative of differential of exponential on $SO(3)$): For given differential of exponential map, dexp_ξ, with exponential coordinates $\xi \in \mathbb{R}^3$ and its time derivative $\dot{\xi}$, the time derivative of differential of exponential map on $SO(3)$ is given by

$$\frac{d}{dt}\mathrm{dexp}_\xi = C_\xi(\dot{\xi}) \tag{34}$$

$$\frac{d}{dt}\mathrm{dexp}_\xi^{-1} = D_\xi(\dot{\xi}). \tag{35}$$

Lemma 4 (Time-derivative of differential of exponential on $SE(3)$): For given differential of exponential map, dexp_λ, with exponential coordinates $\lambda = (\eta^T, \xi^T)^T \in \mathbb{R}^6$ and its time derivative $\dot{\lambda}$, the time-derivative of differential of exponential map on $SE(3)$ is given by

$$\frac{d}{dt}\mathrm{dexp}_\lambda = \begin{bmatrix} C_\xi(\dot{\xi}) & \dot{C}_{\xi,\dot{\xi}}(\eta,\dot{\eta}) \\ 0_3 & C_\xi(\dot{\xi}) \end{bmatrix} \tag{36}$$

$$\frac{d}{dt}\mathrm{dexp}_\lambda^{-1} = \begin{bmatrix} D_\xi(\dot{\xi}) & \dot{D}_{\xi,\dot{\xi}}(\eta,\dot{\eta}) \\ 0_3 & D_\xi(\dot{\xi}) \end{bmatrix} \tag{37}$$

where $\dot{C}_{\xi,\dot{\xi}}(y,\dot{y})$ and $\dot{D}_{\xi,\dot{\xi}}(y,\dot{y})$ are given by

$$\begin{aligned} \dot{C}_{\xi,\dot{\xi}}(y,\dot{y}) = {} & \frac{\beta}{2}(\lceil\dot{y}\rceil - \zeta(y)\lceil\xi\rceil) \\ & + \Gamma_1(\lceil\dot{y},\xi\rceil + \lceil y,\dot{\xi}\rceil + \zeta(y)\lceil\xi\rceil) \\ & + \Gamma_2(\delta_1(y,\dot{y}) + \zeta(y)\lceil\xi\rceil^2) \\ & + \Gamma_3\delta_2(y,\dot{y}) \end{aligned} \tag{38}$$

$$\begin{aligned} \dot{D}_{\xi,\dot{\xi}}(y,\dot{y}) = {} & -\frac{1}{2}\lceil\dot{y}\rceil + \frac{2}{\|\xi\|^2}\frac{(1-\frac{\gamma}{\beta})}{\|\xi\|^2}\zeta(y)\lceil\xi\rceil^2 \\ & + \Gamma_4(\lceil\dot{y},\xi\rceil + \lceil y,\dot{\xi}\rceil) \\ & + \Gamma_5\delta_3(y,\dot{y}) \end{aligned} \tag{39}$$

in which $\zeta(y) = \xi^T y\frac{\xi^T\dot{\xi}}{\|\xi\|^2}$ and Γ_is are

$$\Gamma_1(\xi) = \frac{1-\alpha}{\|\xi\|^2} \tag{40a}$$

$$\Gamma_2(\xi) = \frac{\alpha-\beta}{\|\xi\|^2} \tag{40b}$$

$$\Gamma_3(\xi) = \frac{1}{\|\xi\|^2}\left(\frac{\beta}{2} - \frac{3(1-\alpha)}{\|\xi\|^2}\right) = \frac{\frac{\beta}{2} - 3\Gamma_1(\xi)}{\|\xi\|^2} \tag{40c}$$

$$\Gamma_4(\xi) = \frac{1-\gamma}{\|\xi\|^2} = -\frac{\Gamma_2(\xi)}{\beta} \tag{40d}$$

$$\Gamma_5(\xi) = \frac{1}{\|\xi\|^2}\frac{\frac{1}{\beta}+\gamma-2}{\|\xi\|^2} = \frac{\Gamma_1(\xi) + 2\Gamma_2(\xi)}{\|\xi\|^2\beta}. \tag{40e}$$

The other coefficients δ_is are

$$\delta_1(\eta,\dot{\eta}) = \xi^T\eta\lceil\dot{\xi}\rceil + \xi^T\dot{\xi}\lceil\eta\rceil + (\delta_0(\eta,\dot{\eta}) - 4\zeta(\eta))\lceil\xi\rceil \tag{41a}$$

$$\delta_2(\eta,\dot{\eta}) = \xi^T\eta\lceil\xi,\dot{\xi}\rceil + \xi^T\dot{\xi}\lceil\xi,\eta\rceil + (\delta_0(\eta,\dot{\eta}) - 5\zeta(\eta))\lceil\xi\rceil^2 \tag{41b}$$

$$\delta_3(\eta,\dot{\eta}) = \xi^T\eta\lceil\xi,\dot{\xi}\rceil + \xi^T\dot{\xi}\lceil\xi,\eta\rceil + (\delta_0(\eta,\dot{\eta}) - 3\zeta(\eta))\lceil\xi\rceil^2 \tag{41c}$$

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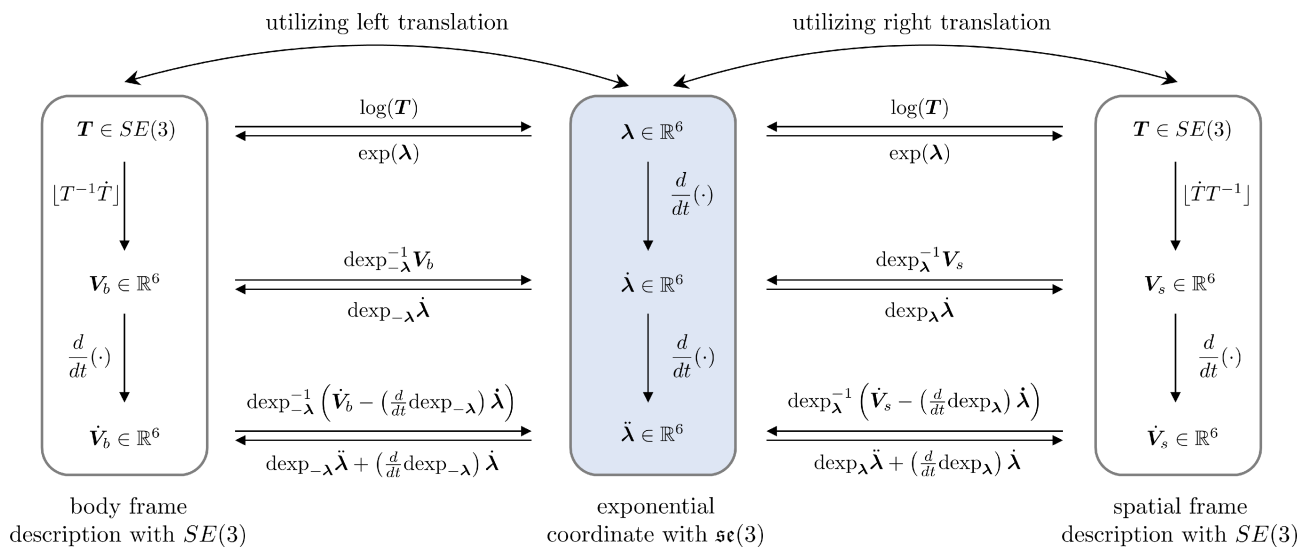

Fig. 2. Commutative mapping between Lie group $SE(3)$ and Lie algebra $\mathfrak{se}(3)$ is illustrated. The body twist and its derivative are utilized in the left translation, while the spatial twist and its derivative are in the right translation. They are linked with the time derivative of the exponential coordinate through the left- and right-trivialized differential of the exponential map, respectively. Note that the inverse image of the exponential map is denoted by $\log(T)$, and $\lfloor\cdot\rfloor$ denotes the inverse operation of the ceiling form $\lceil\cdot\rceil$.

where $\delta_0(\eta, \dot{\eta}) = \dot{\xi}^T\eta + \xi^T\dot{\eta}$. If $\xi = 0_{3\times1}$, they reduce to

$$\dot{C}_{0,\dot{\xi}}(y, \dot{y}) = \frac{1}{2}\lceil \dot{y} \rceil + \frac{1}{6}\lceil \dot{\xi}, y \rceil \tag{42}$$

$$\dot{D}_{0,\dot{\xi}}(y, \dot{y}) = -\frac{1}{2}\lceil \dot{y} \rceil + \frac{1}{12}\lceil \dot{\xi}, y \rceil. \tag{43}$$

The above four Lemmas complete the commutativity between the Lie group and the Lie algebra illustrated in Fig. 2. In other words, the homogeneous transformation, body twist, and twist derivative are connected to the corresponding exponential coordinates using the proposed lemmas. On the contrary, the exponential coordinate, its derivative, and the second derivative are lifted to the homogeneous transformation, body twist, and twist derivative using the lemmas. Again, note that the right-trivialized version of the differential of the exponential map gives the relation between the spatial twist and the time derivative of the exponential coordinate.

III. Impedance Control Design in $SE(3)$

Based on the commutative mapping between the Lie group $SE(3)$ and the Lie algebra $\mathfrak{se}(3)$, we can formulate the impedance control design strategy.

A. Lie Group Impedance Control

If we consider the impedance design at the end-effector coordinate frame, we can assign two frames: the current end-effector frame $\{b\}$ and the desired equilibrium frame $\{d\}$, as shown in Fig. 3. Here, the current end-effector frame denotes the body frame at the current time step, and the desired equilibrium frame denotes the coordinate frame, giving a virtual equilibrium position with zero interaction force. Therefore, the frame $\{b\}$ converges to the $\{d\}$ in free space and deviates from $\{d\}$ depending on the desired impedance relation when the physical interaction

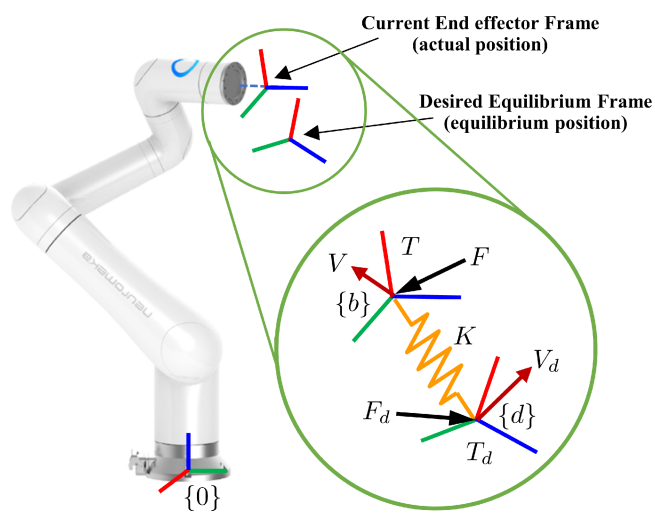

Fig. 3. Impedance illustration in the robot manipulator between the current end-effector frame $\{b\}$ and the desired equilibrium frame $\{d\}$ is shown. The robot base frame is with spatial frame $\{0\}$.

occurs. To design the impedance relation between two frames $\{b\}$ and $\{d\}$, first, we define the relative transformation matrix in the Lie group $SE(3)$ as

$$\tilde{T} = T^{-1}T_d \tag{44}$$

where T and T_d are the current and desired transformation matrices, respectively. Note that $T = {}^0T_b$ and $T_d = {}^0T_d$ are represented in spatial frame $\{0\}$. Then, the relative transformation matrix $\tilde{T} = T^{-1}T_d = {}^bT_0\,{}^0T_d = {}^bT_d$ is the transformational displacement between two frames $\{b\}$ and $\{d\}$, represented in the current end-effector frame $\{b\}$. Here, each body twist can be calculated; current body twist $\lceil V \rceil = T^{-1}\dot{T}$ and desired body twist $\lceil V_d \rceil = T_d^{-1}\dot{T}_d$, respectively. Now, we have two options

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to compute the relative twist. If we utilize the right translation[6] $\lceil \tilde{V} \rceil = \dot{\tilde{T}}\tilde{T}^{-1}$, the resulting relative twist is represented in the current end-effector frame as follows:

$$\tilde{V} = \mathrm{Ad}_{\tilde{T}} V_d - V. \tag{45}$$

Here, (45) implies twist calculation after performing the coordinate transformation of the desired body twist V_d, which is represented in $\{d\}$ frame, into the current end-effector frame $\{b\}$ using the adjoint transformation $\mathrm{Ad}_{\tilde{T}}$. Thus, the relative twist $\tilde{V}$ is finally represented in the $\{b\}$ frame. In contrast, if we utilize the left translation $\lceil \tilde{V} \rceil = \tilde{T}^{-1}\dot{\tilde{T}}$, the resulting relative twist is represented in the desired equilibrium frame as follows:

$$\tilde{V} = V_d - \mathrm{Ad}_{\tilde{T}}^{-1} V. \tag{46}$$

Between them, we choose the relative twist represented in the current end-effector frame as described in (45).

Since the relative twist is obtained through the right translation for the current end-effector frame description, the equivalent exponential coordinates in the associated Lie algebra $\mathfrak{se}(3)$ are defined using (23) as

$$\lambda = \log(\tilde{T}) \tag{47}$$

$$\dot{\lambda} = \mathrm{dexp}_{\lambda}^{-1}\tilde{V} \tag{48}$$

$$\ddot{\lambda} = \mathrm{dexp}_{\lambda}^{-1}\dot{\tilde{V}} + \frac{d}{dt}\mathrm{dexp}_{\lambda}^{-1}\tilde{V}. \tag{49}$$

On the one hand, the external wrench, which is associated with the relative twist $\tilde{V}$, is calculated as[7]

$$\tilde{F} = \mathrm{Ad}_{\tilde{T}}^{-T} F_d - F \tag{50}$$

where F_d is a desired wrench and F is a (body) wrench measured by F/T sensor. Notice that the body wrench is defined as the linear force $f \in \mathbb{R}^3$ and the angular moment $n \in \mathbb{R}^3$, which are applied to the body, to represent the generalized force exerted on a rigid body with respect to the body frame. It is expressed as a six-dimensional vector representation, i.e., $F = (f^T, n^T)^T \in \mathbb{R}^6$.

To impose the dynamic behavior on the robot, we derive the impedance relation based on the energy consideration of the system, especially with the potential energy function computed by exponential coordinates. If we design the positive definite desired inertia matrix $A = \mathrm{diag}(mI_3, \mathcal{I}) \in \mathbb{R}^{6\times 6}$, where m is the desired mass and $\mathcal{I} \in \mathbb{R}^{3\times 3}$ is the moment of inertia, the kinetic energy of the system is given by

$$\mathcal{K} = \frac{1}{2}\tilde{V}^T A \tilde{V}. \tag{51}$$

Then, we take the time derivative of (51), which is given by

$$\dot{\mathcal{K}} = \dot{\tilde{V}}^T A \tilde{V} = (A\dot{\tilde{V}})^T \tilde{V}. \tag{52}$$

It gives the inertial wrench $A\dot{\tilde{V}}$, contributing to the impedance relation. In addition, a dissipative wrench can be considered as $D\tilde{V}$ for the positive definite desired damping matrix $D \in \mathbb{R}^{6\times 6}$. For an elastic wrench, if we consider the positive definite desired stiffness matrix $K \in \mathbb{R}^{6\times 6}$, the potential energy of the system is given by

$$\mathcal{U} = \frac{1}{2}\lambda^T K \lambda. \tag{53}$$

Taking a time derivative of (53), then

$$\begin{aligned}\dot{\mathcal{U}} &= \lambda^T K \dot{\lambda} \\ &= \lambda^T K(\mathrm{dexp}_{\lambda}^{-1}\tilde{V}) = (\mathrm{dexp}_{\lambda}^{-T} K\lambda)^T \tilde{V}.\end{aligned} \tag{54}$$

Here, the relation (48) is used. So, the elastic wrench contributing to the impedance relation is obtained as $\mathrm{dexp}_{\lambda}^{-T} K\lambda$. Therefore, the desired impedance model in the Lie group $SE(3)$ can be designed as

$$A\dot{\tilde{V}} + D\tilde{V} + \mathrm{dexp}_{\lambda}^{-T} K\lambda = \tilde{F} \tag{55}$$

where again A, D, and K are desired inertia, damping, and stiffness matrices in Lie group $SE(3)$, respectively, and $\tilde{F}$ is the wrench above associated with $\tilde{V}$ defining a virtual work $\tilde{V}^T\tilde{F}$.

Now, we want to design the controller in $\mathfrak{se}(3)$ to take advantage of the Lie algebra, which is a vector space. The impedance relation in the Lie group $SE(3)$ must be transformed to the corresponding impedance relation in the Lie algebra $\mathfrak{se}(3)$. Using the relations (47), (48), and (49), the exponential coordinates are already computed. The remaining variable is a wrench component associated with exponential coordinates in the Lie algebra. Now, the question arises: How can we obtain this entity, which is the conversion of $\tilde{F}$ to be associated with an exponential coordinate? We call it an equivalent effective wrench that defines virtual work together with $\dot{\lambda}$ in the associated Lie algebra, and it is expressed as γ. Using the virtual work theorem (virtual power conservation), the calculation procedure is as follows[8]:

$$\begin{aligned}\dot{\lambda}^T\gamma &= \tilde{V}^T\tilde{F} \\ (\mathrm{dexp}_{\lambda}^{-1}\tilde{V})^T\gamma &= \tilde{V}^T\tilde{F} \\ \tilde{V}^T(\mathrm{dexp}_{\lambda}^{-T}\gamma) &= \tilde{V}^T\tilde{F} \\ \therefore \quad \gamma &= \mathrm{dexp}_{\lambda}^{T}\tilde{F}.\end{aligned} \tag{56}$$

Therefore, the impedance equation in $SE(3)$ can be converted to $\mathfrak{se}(3)$, which defines the error dynamics in $\mathfrak{se}(3)$, as

$$A_\lambda\ddot{\lambda} + D_\lambda\dot{\lambda} + K_\lambda\lambda = \gamma \tag{57}$$

where the conversion relations of impedance parameters between $SE(3)$ and $\mathfrak{se}(3)$ are calculated as follows:

$$A_\lambda = \mathrm{dexp}_{\lambda}^{T} A \mathrm{dexp}_{\lambda} \tag{58a}$$

[6]From the commutative mapping described in Fig. 2, we can understand that the right translation property preserves the "representing frame" (upper left index $\{b\}$). In contrast, the left translation property preserves the "represented frame" (lower right index $\{d\}$) for a given transformation matrix $\tilde{T} = T^{-1}T_d = {}^bT_d$.

[7]For the desired body twist V_d, we use $\mathrm{Ad}_{\tilde{T}}$ for coordinate transformation from the desired to the current frame. Since the body wrench is dual to the body twist, the desired wrench F_d takes the inverse transpose operation of the adjoint transform, i.e., $\mathrm{Ad}_{\tilde{T}}^{-T}$.

[8]The derivation procedure of the equivalent effective wrench resembles the relation between joint torque τ and body wrench F, i.e., $\tau = J^T F$ with body Jacobian matrix J. This relation is also obtained through the duality condition of the equation $V = J\dot{\theta}$.

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$$D_\lambda = \mathrm{dexp}_\lambda^T \left(D\mathrm{dexp}_\lambda + A\frac{d}{dt}\mathrm{dexp}_\lambda \right) \tag{58b}$$

$$K_\lambda = K. \tag{58c}$$

In other words, the impedance parameters in the Lie group $SE(3)$ can be converted into those of the Lie algebra $\mathfrak{se}(3)$ and vice versa using a commutative mapping. Note that the damping D_λ related to the exponential coordinate may no longer be symmetric due to the nonlinearity of the differential of the exponential map, even though the damping D related to the relative twist in $SE(3)$ is symmetric. That is why the impedance design should be in the Lie group $SE(3)$, not directly in the Lie algebra $\mathfrak{se}(3)$.

The proposed detour strategy that utilizes the Lie algebra element can provide a concise and consistent formulation due to the vector space isomorphism [29], [30]. With the commutative mapping between the Lie group and the Lie algebra shown in Fig. 2, we can perform the controller design in the Lie algebra $\mathfrak{se}(3)$. To achieve the transformed desired impedance behavior in $\mathfrak{se}(3)$, we follow the reference acceleration tracking strategy [30]:

$$\ddot{\lambda} = \ddot{\lambda}_{\mathrm{ref}}. \tag{59}$$

By the impedance relation (57), we have

$$\begin{aligned} \ddot{\lambda}_{\mathrm{ref}} &= -A_\lambda^{-1} D_\lambda \dot{\lambda} - A_\lambda^{-1} K_\lambda \lambda + A_\lambda^{-1}\gamma \\ &= -K_V \dot{\lambda} - K_P \lambda + K_F \gamma. \end{aligned} \tag{60}$$

The idea of reference acceleration tracking is that if the acceleration $\ddot{\lambda}$ faithfully follows $\ddot{\lambda}_{\mathrm{ref}}$ using the control, we can achieve the desired dynamics (57). The impedance parameters act as gain matrices and are simplified as follows:

$$K_V = \mathrm{dexp}_\lambda^{-1} \left(A^{-1} D\mathrm{dexp}_\lambda + \frac{d}{dt}\mathrm{dexp}_\lambda \right) \tag{61a}$$

$$K_P = \mathrm{dexp}_\lambda^{-1} A^{-1} \mathrm{dexp}_\lambda^{-T} K \tag{61b}$$

$$K_F = \mathrm{dexp}_\lambda^{-1} A^{-1} \mathrm{dexp}_\lambda^{-T}. \tag{61c}$$

Now, we need to return to the joint space to actually control the robot joints through $SE(3)$. To get the reference acceleration in Lie group $SE(3)$, we use the time derivative of the relative twist (45) as

$$\dot{\tilde{V}} = \mathrm{Ad}_{\tilde{T}} \dot{V}_d + \mathrm{ad}_{\tilde{V}} \mathrm{Ad}_{\tilde{T}} V_d - \dot{V}. \tag{62}$$

Equation (49) can be rewritten as

$$\dot{\tilde{V}} = \mathrm{dexp}_\lambda \ddot{\lambda} + \frac{d}{dt}\mathrm{dexp}_\lambda \dot{\lambda} \tag{63}$$

by using the following relation:

$$\mathrm{dexp}_\lambda \frac{d}{dt}\mathrm{dexp}_\lambda^{-1}\mathrm{dexp}_\lambda = -\frac{d}{dt}\mathrm{dexp}_\lambda.$$

Equating (62) and (63) using $\ddot{\lambda} = \ddot{\lambda}^{\mathrm{ref}}$, we can calculate $\dot{V}$ as a reference twist derivative $\dot{V}_{\mathrm{ref}}$

$$\dot{V}_{\mathrm{ref}} = \mathrm{Ad}_{\tilde{T}} \dot{V}_d - \mathrm{dexp}_\lambda \ddot{\lambda}_{\mathrm{ref}} + \mathrm{ad}_{\tilde{V}} \mathrm{Ad}_{\tilde{T}} V_d - \frac{d}{dt}\mathrm{dexp}_\lambda \dot{\lambda}$$

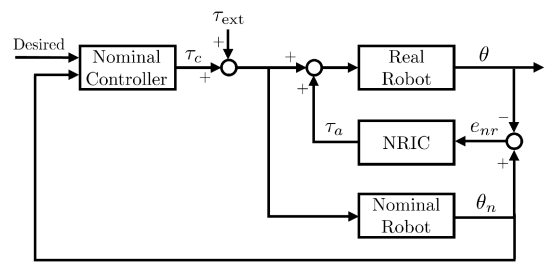

Fig. 4. Robust impedance control structure using NRIC [5], [36]. In the figure, the nominal controller block matches the proposed Lie group impedance controller in (65).

$$= \mathrm{Ad}_{\tilde{T}} \dot{V}_d - \mathrm{dexp}_\lambda \ddot{\lambda}_{\mathrm{ref}} + \mathrm{ad}_{\tilde{V}} V - \frac{d}{dt}\mathrm{dexp}_\lambda \dot{\lambda} \tag{64}$$

since $\mathrm{ad}_{\tilde{V}} \tilde{V} = \mathrm{ad}_{\tilde{V}}(\mathrm{Ad}_{\tilde{T}} V_d - V) = 0_{6\times 1}$. Now, the control torque is obtained through inverse dynamic control as

$$\tau_c = M(\theta)\ddot{\theta}_{\mathrm{ref}} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta) - J^T(\theta) F \tag{65}$$

where $M(\theta)$, $C(\theta, \dot{\theta})$, and $g(\theta)$ represent the joint space inertia, Coriolis matrices, and the gravity vector of the robot, respectively. The reference joint acceleration is computed by

$$\ddot{\theta}_{\mathrm{ref}} = J(\theta)^{-1} \left(\dot{V}_{\mathrm{ref}} - \dot{J}(\theta, \dot{\theta})\dot{\theta} \right) \tag{66}$$

where $J(\theta)$ is Jacobian matrix. Under the control input computed in (65), the closed-loop system becomes the desired impedance relation described in (55). Thus, the desired impedance is realized through the proposed Lie group impedance controller.

B. Robust Impedance Control Structure

From an implementation perspective, it is well known that the traditional (explicit) impedance control law is affected by the model uncertainty and disturbances since it requires accurate model parameters to achieve the closed-loop dynamics as designed [11], [37]. It can also happen in the proposed method described in the previous subsection due to the inverse dynamic control strategy in (65) and (66). Hence, the control performance degrades under model uncertainties and disturbances in real robot experiments. This often makes it difficult to verify the proposed six-DoF impedance control framework in a real-world scenario. To achieve robust performance, a nonlinear robust internal loop compensator (NRIC) illustrated in Fig. 4 is utilized [5], [36]. The NRIC guarantees that the real robot state asymptotically converges to the nominal robot even under model uncertainties and disturbances. Therefore, the nominal performance, which realizes the desired impedance behavior, can be obtained. In addition to τ_c in (65), the auxiliary torque input τ_a to make the real robot follow the nominal robot is added through the PID-type position controller as

$$\tau_a = K'_\gamma (\dot{e}_{nr} + K'_p e_{nr} + K'_i \int e_{nr}) \tag{67}$$

where K'_γ, K'_p, and K'_i are the PID control gain matrices, and $e_{nr} = \theta_n - \theta$ is the joint position error between the nominal and the real robot. Due to the NRIC structure, the Lie group

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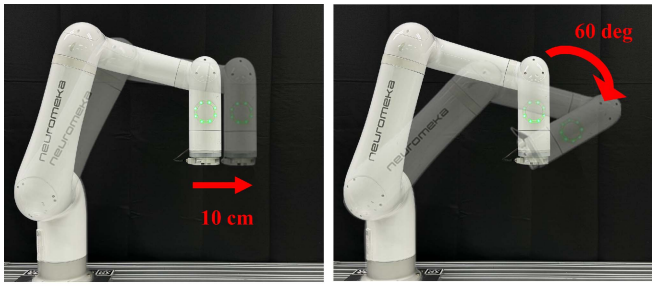

(a) (b)

Fig. 5. Experiment description for (a) translational (b) rotational dynamics rendering in Section IV-A: In the experiment, the regulation scenarios are considered. Therefore, the step input is entered, and we observe the behaviors. The desired step position and orientation are given as 10 cm and 60 deg, respectively.

impedance controller can achieve robust performance, as it enhances the robustness of any type of controller.

IV. Experimental Results

To evaluate the proposed impedance control, experiments were performed on a 6-DoF robot manipulator (Indy7, Neuromeka). For hardware implementation, the robot is equipped with a 6-DoF F/T sensor (RFT80-6A01-A, Robotus) to measure the external wrench applied to the end-effector. In particular, the 3D-printed tool was attached to the end-effector to measure the external wrench through the F/T sensor for interaction. During the experiment, control frequency was 4 kHz.

A. Rendering Desired Dynamics

First, we check whether the proposed Lie group impedance control can accurately render the desired inertia as specified by the impedance model. The rendering of dynamics is divided into translation and rotation, as illustrated in Fig. 5. Then, we perform position and orientation regulation tasks to observe the desired translational and rotational dynamic behaviors. In the first experiment [see Fig. 5(a)], we fix the desired translational damping D_t and stiffness K_t as

$$D_t = 20I_3, \;\; K_t = 20I_3 \tag{68}$$

and abruptly changes the desired position. Under the proposed Lie group impedance control, we design three different desired translational inertia as

$$A_{t,1} = 0.5I_3, \;\; A_{t,2} = 5I_3, \;\; A_{t,3} = 100I_3. \tag{69}$$

Fig. 6(a) shows the result of the first experiment as we alter the desired inertia; the dynamic behavior changes under the proposed Lie group impedance control. The robot shows under damped motion as the desired inertia increases [see the plots in Fig. 6(a)]. Note that, for the usual second-order closed-loop error dynamics $A_t\ddot{e} + D_t\dot{e} + K_t e = 0$ with $e = p_d - p$, the desired dynamic behavior is usually divided into the over damped, critical-damped, and under damped systems, depending on the sign of the term $D_t^2 - 4A_tK_t$. Since we fix $D_t = 20I_3$ and $K_t = 20I_3$, the apparent inertia $A_{t,1}$, $A_{t,2}$, and $A_{t,3}$ denote the

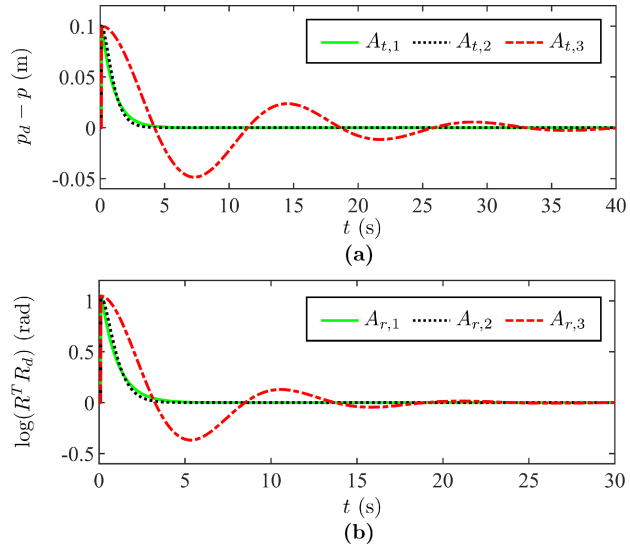

Fig. 6. Experimental result of desired dynamics rendering is shown under the same damping and stiffness, with different inertia. (a) Translational dynamics, and (b) rotational dynamics. The green, black, and red lines render the over damped, critical-damped, and under damped systems, respectively. As the desired inertia increases, the robot shows slow convergence with overshoot. It implies that the desired dynamic behavior (system response) is realized by rendering the different desired inertia.

over damped, critical-damped, and under damped systems, respectively. This intuition directly matches the experiment result. In the second experiment [see Fig. 5(b)], we want to observe the rotational dynamics. We fix the desired rotational damping and stiffness as

$$D_r = 10I_3, \;\; K_r = 10I_3 \tag{70}$$

and abruptly changes the desired orientation. For the case of the proposed Lie group impedance control, we design three different desired rotational inertia as

$$A_{r,1} = 0.5I_3, \;\; A_{r,2} = 2.5I_3, \;\; A_{r,3} = 25I_3. \tag{71}$$

Similarly to the first experiment, the apparent inertia $A_{r,1}$, $A_{r,2}$, and $A_{r,3}$ denote the over damped, critical-damped, and under damped systems, respectively, as shown in Fig. 6(b). As we increase the desired inertia, the robot shows slow convergence with overshoot [see the plots in Fig. 6(b)]. Therefore, the proposed Lie group impedance control can render the desired dynamic behavior with inertia reshaping.

B. Comparison to Existing Methods

Now, we will compare the proposed Lie group impedance control with other existing methods that render the desired inertia, damping, and stiffness for the six-DoF rigid body motion.

1) Resolving Topological Obstruction Issue in Quaternion: A unit quaternion is a nonminimal representation that describes the rigid body orientation. Since it resolves the representation singularity of the Euler angle, it has been used to design the orientational impedance control [11], [12], [13]. However, it is reported that those controls suffer from the topological obstruction issue when the imaginary part of the unit quaternion is used to describe the orientational displacement [14]. The topological

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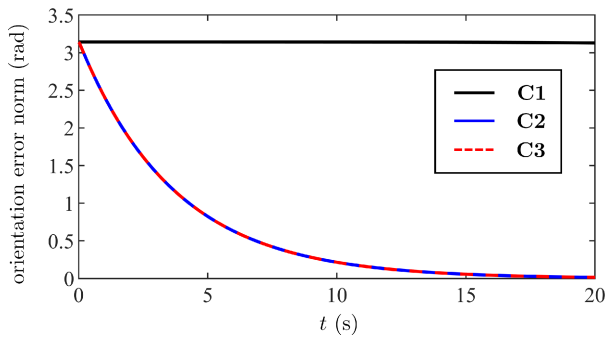

Fig. 7. Orientation error norm plot for the experiment checking topological obstruction. When the imaginary part of the unit quaternion is used, the system keeps the current orientation even if the desired orientation is nonzero (a half-turn). This is an undesirable situation.

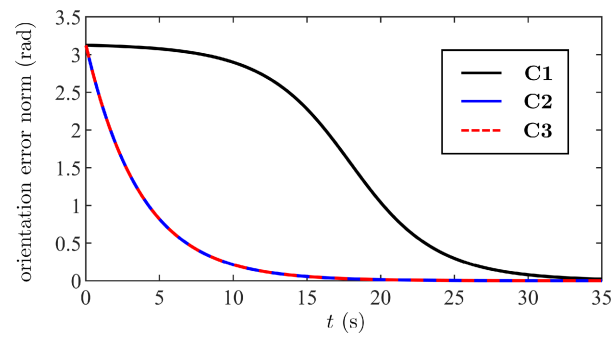

Fig. 8. Orientation error norm plot for the experiment checking the behavior under the Lie group and Lie algebra potential energy function.

obstruction causes the controller to be trapped in unstable equilibrium sets: the rotational axis is parallel to the eigenvectors of the stiffness matrix when the orientational displacement is $\pi\,\mathrm{rad}$ [15]. For this reason, we set the desired orientation as $\pi\,\mathrm{rad}$ around the z-axis from the current orientation, that is, $R_d = R_0\mathrm{Rot}(\hat{z}, \pi)$ where R_0 is the initial orientation of the end-effector[9], and observe the behavior. During the experiments, the desired impedance parameters are chosen as

$$A = 4I_6,\ D = 150I_6,\ K = 40I_6. \tag{72}$$

The types of controllers corresponding to orientational displacement are described as follows:

1) C1: unit quaternion imaginary part
2) C2: unit quaternion logarithm map
3) C3: proposed Lie group impedance control.

C1 is a well-known orientational impedance control design utilizing the imaginary part of the unit quaternion [11], but it suffers from topological obstruction. C2 is a six-DoF impedance control based on the dual quaternion logarithm map in [15]; the orientation part is described as a unit quaternion logarithm map, and it was shown that the topological obstruction can be resolved. C3 is the proposed Lie group impedance control design. Therefore, in this experiment, we want to check whether the proposed Lie group impedance control can resolve the topological obstruction as the dual quaternion logarithm map did.

In Fig. 7, C1, which utilizes the imaginary part of the unit quaternion, suffers from the topological obstruction issue. The controller does not generate the control torque, even though the orientation error exists as $\pi\,\mathrm{rad}$, and the end-effector does not rotate and converge to the desired orientation. However, C2, which utilizes the logarithm map of the unit quaternion, can resolve the issue; it converges to the desired orientation. Similarly, the proposed Lie group impedance controller converges to the desired orientation without the topological obstruction issue. Thus, even if the proposed method utilizes minimal representation, it can resolve the topological obstruction, as the dual quaternion logarithm method utilizing nonminimal representation does.

[9]In this article, the notation $\mathrm{Rot}(\hat{x}, \pi/4)$ implies x-axis rotation by $\pi/4\,\mathrm{rad}$.

2) Convergence Under Lie Group and Lie Algebra Potential Energy Function: Next, we want to check the behavior under the Lie group and the Lie algebra potential energy function formulation. As mentioned in Section I, the related approaches in [19], [20], [21], [22], [25], and [26] utilize the potential energy function directly in the Lie group, so different behavior will occur compared to the proposed method, which utilizes the potential energy function in the Lie algebra. In fact, for the Lie group potential energy function, the rotation angle has been reported to be not linear to the control input generated by the rotational spring [27], and the control input vanishes when the rotation angle becomes $\pi\,\mathrm{rad}$. Note that the mentioned works with the Lie group energy function are realized by PD-based impedance control implementations. They cannot reshape inertia with their current formulation. So, it is difficult or even unfair to compare the proposed method. Fortunately, it is revealed that the impedance control design using the imaginary part of the unit quaternion shares the same rotational potential energy formulation in the Lie group [38] and is exactly equivalent to the work in [19], [20], [21], and [22]. Therefore, we judge that it is enough to consider the case of the unit quaternion imaginary part, which was already introduced in the previous experiment, to check the behavior under the Lie group potential energy function approach.

For the comparison between the Lie group and the Lie algebra potential energy functions, we set the desired orientation near $\pi\,\mathrm{rad}$ around the z-axis from the current orientation, that is, $R_d = R_0\mathrm{Rot}(\hat{z}, \frac{179}{180}\pi)$ where R_0 is the initial orientation of the end-effector. The desired impedance parameters are chosen as the same in (72), and we observe the behavior during convergence. The same controllers, C1, C2, and C3, are used as those in the previous experiment.

Fig. 8 shows the experimental result. C1, which utilizes the Lie group potential energy function, shows slow convergence to the desired orientation. In fact, the control input calculated under the Lie group potential energy formulation has a nonlinear relation to the rotation angle, and it vanishes at rotation about $\pi\,\mathrm{rad}$. Thus, the control input is very small near the $\pi\,\mathrm{rad}$. It causes slow convergence behavior as it starts. Instead, C2 and C3 utilize the unit quaternion logarithmic variable and exponential coordinate, respectively. Those are based on the Lie algebra potential energy function formulation. Therefore, both cases show exponentially

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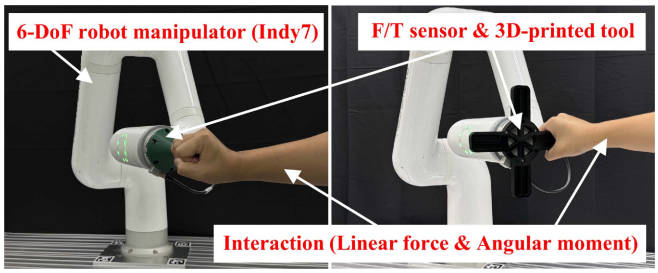

Fig. 9. Experimental description for the interaction test in Section IV-C. (Left) Interaction with linear force. (Right) Interaction with angular moment.

decaying error convergence for the orientation part. Note that C2 is based on the nonminimal representation, while C3, which is our proposed method, is based on the minimal representation.

C. Experiments Under External Wrench

Lastly, we confirm that the proposed Lie group impedance control can realize the desired impedance, including zero stiffness realization. For this purpose, the desired impedance parameters are chosen as follows:

$$\begin{aligned} A &= \mathrm{diag}(4,4,4,1,1,1) \\ D &= \mathrm{diag}(40,40,40,10,10,10) \\ K &= \mathrm{diag}(100,0,2000,0,20,60) \end{aligned} \tag{73}$$

where the first three components are for the translation of the end-effector and the last three components are for the rotation. Note that the translational stiffness on the y-axis and the rotational stiffness on the x-axis are zero, respectively. The end-effector should freely move along the y-direction and rotate along the x-direction without converging to its original position and orientation, respectively. In contrast, the end-effector should maintain the same position in the z-direction due to the high translational stiffness. The overview of the experimental setup is shown in Fig. 9. During the experiment, the human operator applied an external wrench through the F/T sensor mounted at the end-effector for interaction.

The experimental results are shown in Figs. 10 and 11. Under the proposed Lie group impedance control, the positional displacement along the y-axis occurs during the interaction. Still, it does not return to zero in the absence of the interaction force [see the second row in Fig. 10]. This means that the end-effector moves freely along the y-axis and does not converge to its original position; zero translational stiffness is realized along the y-axis. Instead, the positional displacement along the z-axis maintains a zero position. Even under the large z-directional force around $50\,\mathrm{N}$ [red lines in the third row in Fig. 10], the robot moves little as $0.025 = 50/2000\,\mathrm{m}$ and maintains its original position along the z-direction. It strictly realizes the high translational stiffness on the z-axis. Similarly, the orientational displacement along the x-axis occurs and does not return to zero [see the first row in Fig. 11]. This implies that the end-effector rotates freely along the x-axis and does not converge to its original orientation; zero rotational stiffness along the x-axis is realized.

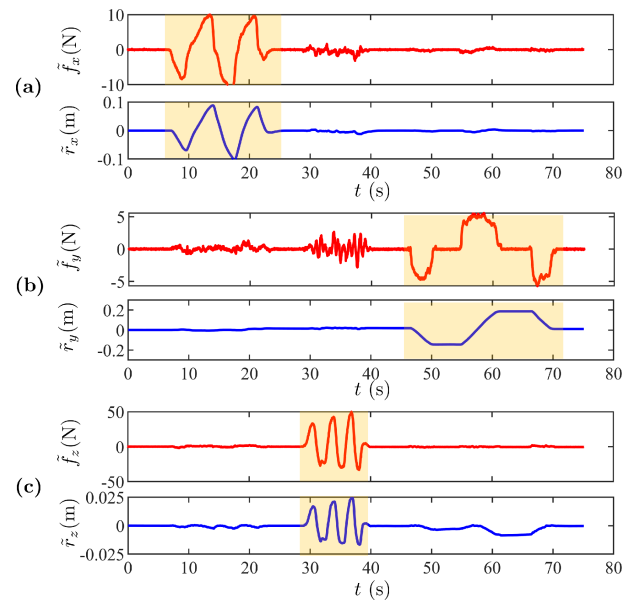

Fig. 10. Applied linear force and positional displacement plots of the robot end-effector. (a), (b), and (c) Applied forces and positional displacements in x-, y-, and z-directions w.r.t. the end-effector coordinate frame, respectively. Here, $\tilde{r}$ in the graphs means the translation part in $\tilde{T} = T^{-1}T_d$, i.e., $\tilde{r} = R^T(p_d - p)$, and the $\tilde{f}$ means the applied force that is the linear part in $\tilde{F} = \mathrm{Ad}_{\tilde{T}}^{-T} F_d - F$. The yellow-shaded region denotes the time interval of interaction.

Furthermore, Fig. 12 shows the desired stiffness realization by plotting the force-displacement graphs except for the zero stiffness cases. Following the desired stiffness design in (73), the blue dashed line shows the relevant desired stiffness value, and the red dotted line denotes the robot behavior during the interaction test. The black line indicates the ideal behavior calculated by the desired impedance equation in (55). Therefore, it is confirmed that the high translational stiffness along the z-axis is realized, and the different stiffness values for each axis, including position and orientation, are realized. Thus, the proposed impedance control can realize the desired impedance with different stiffness implementations.

V. Discussion

A. Singularity Issues in Minimal Representation Approaches and the Proposed Method

In classical approaches, minimal representation, such as the Euler angle, is used to define orientational displacement but suffers from representation singularity. Alternatively, the relative Euler angle can mitigate the representation singularity by extracting the Euler angle from the relative orientation [11], [39]. However, this approach does not completely eliminate the issue.

The proposed six-DoF impedance control can achieve a singularity-free formulation with minimal representation. This is the result of using the exponential coordinate, which exploits the Lie group $SE(3)$ and the Lie algebra $\mathfrak{se}(3)$. However, it is noted that the maps $\exp(\cdot)$ and $\log(\cdot)$ are local diffeomorphisms. The exponential coordinate ξ, which is obtained from the matrix logarithm of the rotation matrix $R \in SO(3)$, is only defined in the region of $\|\xi\| < \pi$, which is called the injectivity radius

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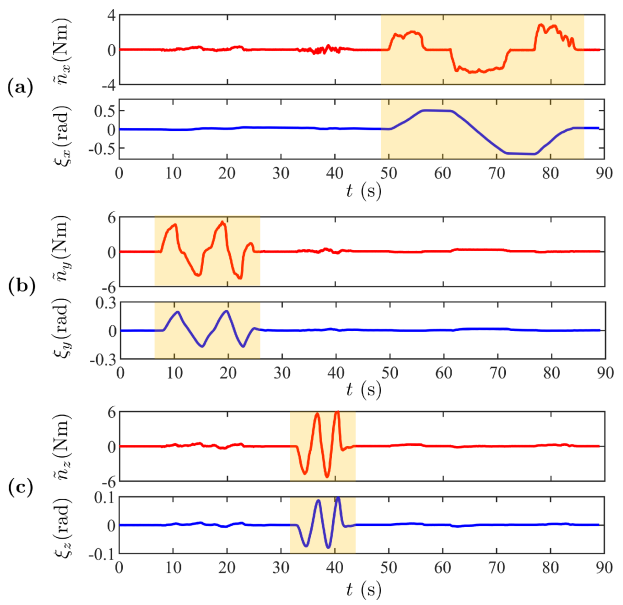

Fig. 11. Applied angular moment and orientational displacement plots of the robot end-effector. (a), (b), and (c) Applied moment and orientational displacements in x-, y-, and z-directions w.r.t. the end-effector coordinate frame, respectively. Here, ξ in the graphs means the rotation part in $\tilde{T} = T^{-1}T_d$, i.e., $\xi = \log(R^T R_d)$, and the $\tilde{n}$ means the applied moment that is the angular part in $\tilde{F} = \mathrm{Ad}_{\tilde{T}}^{-T} F_d - F$. The yellow-shaded region denotes the time interval of interaction.

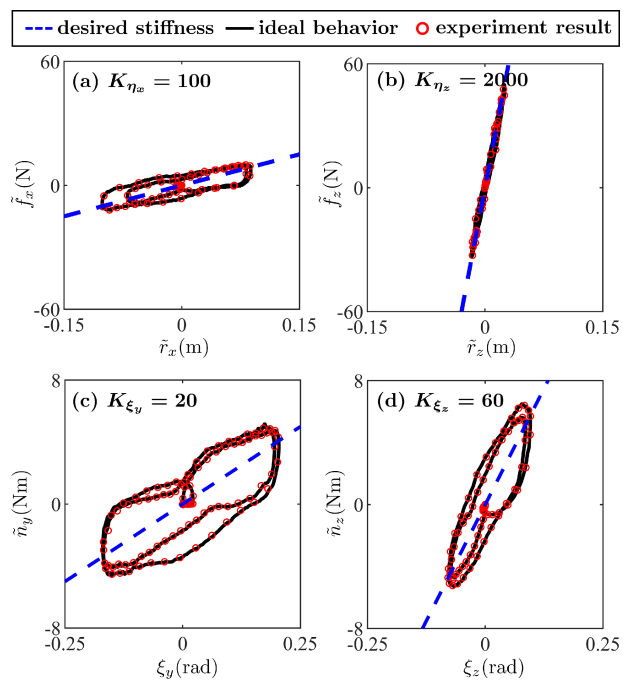

Fig. 12. Force-displacement plot during interaction to check the desired impedance realization. In the figure, we express the desired stiffness matrix $K = \mathrm{diag}(K_{\eta_x}, K_{\eta_y}, K_{\eta_z}, K_{\xi_x}, K_{\xi_y}, K_{\xi_z})$. Each plot corresponds to the interaction period, represented by the yellow-shaded region in Figs. 10 and 11, except for the zero stiffness cases.

(or often referred to as the open ball with radius π). If the relative rotation is over π, then the exponential coordinate ξ is again mapped to $-\pi$. This is a natural intrinsic property in minimal representation [29]. Nevertheless, the proposed approach exploring Lie group theory can avoid the representation singularity commonly occurring in the Euler angle description. An additional comment is that the control based on the dual quaternion logarithm map [15] has the same representation region for orientation as our proposed approach, although it is a nonminimal representation since it uses a switching error function to avoid the unwinding phenomenon that originated from the double-covering property in unit quaternion [40].

As discussed above, the mapping region for the orientational displacement under the proposed method is limited to the injectivity radius π. It is an unavoidable limitation of the proposed method, a consequence of the local nature of the $SO(3)$ group. If we want to overcome this restriction by mapping beyond the region of the injectivity radius, the extended exponential coordinate (EEC) can be utilized [29]. A method to map the exponential coordinate on $SO(3)$ beyond the injectivity radius of π has been proposed by integrating the past trajectory. Fortunately, extending the exponential coordinate on $SE(3)$ beyond the injectivity radius, utilizing the EEC on $SO(3)$ mentioned, is also possible. Since the proposed control framework operates through the commutative map between the Lie group $SE(3)$ and the Lie algebra $\mathfrak{se}(3)$, the proposed method can be extended beyond the injectivity radius if all the commutative maps can be well-defined when the mapping region of ξ is extended (i.e., $\|\xi\| > \pi$). By considering the closed-form formulas described in four lemmas in Section II, all the commutative maps (dexp_ξ, dexp_ξ^{-1}, dexp_λ, $\mathrm{dexp}_\lambda^{-1}$, $\frac{d}{dt}\mathrm{dexp}_\xi$, $\frac{d}{dt}\mathrm{dexp}_\xi^{-1}$, $\frac{d}{dt}\mathrm{dexp}_\lambda$, and $\frac{d}{dt}\mathrm{dexp}_\lambda^{-1}$) are actually well-defined in the region of $\|\xi\| < 2\pi$ because all the coefficients are well-defined in the extended region. At the moment of $\|\xi\| = 2\pi$, they are not defined since the cotangent and cosecant functions originated from the coefficients γ, $1/\beta$, and γ/β, which are not defined. Thus, the proposed method can be extended beyond the injectivity radius until $\|\xi\| = 2\pi$ by introducing the EEC.

B. Stability Analysis of the Proposed Impedance Control

For the Lie group $SE(3)$, it is reported that there is no bi-invariant metric [41], [42], [43]. Since the bi-invariant metric defines the norm (inner product) in the manifold, a direct stability consideration in $SE(3)$ is difficult. Instead, we will suggest a Lyapunov stability for the converted impedance equation in (57) defined in the Lie algebra. Since the associated exponential coordinate can be considered as a six-dimensional vector in Euclidean space, the equation $A_\lambda \ddot{\lambda} + D_\lambda \dot{\lambda} + K_\lambda \lambda = \gamma$ denotes the closed-loop error dynamics in the Euclidean space. Let us consider the following Lyapunov candidate function:

$$H = \frac{1}{2}\dot{\lambda}^T A_\lambda \dot{\lambda} + \frac{1}{2}\lambda^T K_\lambda \lambda. \tag{74}$$

Differentiating the Lyapunov candidate function H along the trajectory of (57) with $\gamma = 0_{6\times 1}$, which is given by

$$\dot{H} = \frac{1}{2}\left(\underbrace{\ddot{\lambda}^T A_\lambda \dot{\lambda}}_{\text{(i)}} + \underbrace{\dot{\lambda}^T \dot{A}_\lambda \dot{\lambda}}_{\text{(iii)}} + \underbrace{\dot{\lambda}^T A_\lambda \ddot{\lambda}}_{\text{(ii)}}\right)$$

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$$+\frac{1}{2}\left(\dot{\lambda}^T K_\lambda \lambda + \lambda^T K_\lambda \dot{\lambda}\right) \tag{75}$$

since $A_\lambda = \mathrm{dexp}_\lambda^T A \mathrm{dexp}_\lambda$ is time-varying parameter for constant and positive definite matrix A, and $K_\lambda = K$ is simply constant and positive definite matrix as in (58). The designated terms are calculated as follows:

$$\text{(i)} = (-D_\lambda \dot{\lambda} - K_\lambda \lambda)^T \dot{\lambda} = -\dot{\lambda}^T D_\lambda^T \dot{\lambda} - \lambda^T K_\lambda \dot{\lambda}$$

$$\text{(ii)} = \dot{\lambda}^T(-D_\lambda \dot{\lambda} - K_\lambda \lambda) = -\dot{\lambda}^T D_\lambda \dot{\lambda} - \dot{\lambda}^T K_\lambda \lambda$$

$$\text{(iii)} = \dot{\lambda}^T \left(\frac{d}{dt}\mathrm{dexp}_\lambda^T A \mathrm{dexp}_\lambda + \mathrm{dexp}_\lambda^T A \frac{d}{dt}\mathrm{dexp}_\lambda\right)\dot{\lambda}.$$

Since $D_\lambda = \mathrm{dexp}_\lambda^T (D\mathrm{dexp}_\lambda + A\frac{d}{dt}\mathrm{dexp}_\lambda)$, the term $\dot{\lambda}^T(D_\lambda + D_\lambda^T)\dot{\lambda}$ can be divided into positive definite part $(*)$ and indefinite part $(**)$

$$\dot{\lambda}^T(D_\lambda + D_\lambda^T)\dot{\lambda} = (*) + (**)$$

where

$$(*) = \dot{\lambda}^T \left(\mathrm{dexp}_\lambda^T D \mathrm{dexp}_\lambda + \mathrm{dexp}_\lambda^T D \mathrm{dexp}_\lambda\right)\dot{\lambda}$$

$$(**) = \dot{\lambda}^T \left(\mathrm{dexp}_\lambda^T A \frac{d}{dt}\mathrm{dexp}_\lambda + \frac{d}{dt}\mathrm{dexp}_\lambda^T A \mathrm{dexp}_\lambda\right)\dot{\lambda}.$$

Here, the indefinite part $(**)$ is exactly the same as the term (iii), so it will be canceled out when we compute the summation of all terms (i) + (ii) + (iii). Thus, (75) becomes

$$\begin{aligned}\dot{H} &= -\frac{1}{2}\dot{\lambda}^T \left(\mathrm{dexp}_\lambda^T D \mathrm{dexp}_\lambda + \mathrm{dexp}_\lambda^T D \mathrm{dexp}_\lambda\right)\dot{\lambda} \\ &= -\dot{\lambda}^T \left(\mathrm{dexp}_\lambda^T D \mathrm{dexp}_\lambda\right)\dot{\lambda}.\end{aligned} \tag{76}$$

Note that the matrix $\mathrm{dexp}_\lambda^T D \mathrm{dexp}_\lambda$ is positive definite if and only if the matrix D is positive definite for a nonsingular matrix dexp_λ. Hence, (76) is negative semidefinite, but using Lasalle's invariance principle, it can be further concluded the asymptotic stability with the equilibrium point $\lambda = 0_{6\times1}$, $\dot{\lambda} = 0_{6\times1}$ for the Lyapunov candidate in (74).

VI. Conclusion

This article proposes a novel framework for designing impedance control in $SE(3)$ that takes advantage of Lie group theory. In the proposed framework, the detour strategy is utilized; the desired impedance is designed in the Lie group $SE(3)$, and the controller is designed in the Lie algebra $\mathfrak{se}(3)$, isomorphic to the Euclidean space. Based on the commutative mapping between the Lie group $SE(3)$ and the Lie algebra $\mathfrak{se}(3)$, the impedance design is directly formulated, maintaining the group structure of $SE(3)$ with minimal representation: six-dimensional exponential coordinates λ. We could observe through the experiment that the proposed Lie group impedance control can render the desired dynamic behavior for six-DoF rigid body motion as a new mass, spring, and damper system. Significantly, applying the Lie group theory ensures a minimal representation that avoids the representation singularities commonly associated with Euler angle descriptions of the orientation part, although the injectivity radius for ξ is intrinsically π for minimal representation approaches. The experiments validated that the proposed Lie group impedance control does not have a topological obstruction in the imaginary part of the unit quaternion. At the same time, the proposed impedance control exhibits exponentially decaying orientational error convergence due to the Lie algebra potential energy function, while for the Lie group potential energy function, the control input is nonlinear to the rotation angle.

For future work, the possible extension of the proposed method beyond the injectivity radius by using the concept of an EEC will be verified through a real-world experiment. Furthermore, the proposed control framework is not limited to applying a robot manipulator. Since it describes the six-DoF rigid body motion in the context of the Lie group $SE(3)$, it can be extended to other robot systems that take the homogeneous transformation matrix as the robot state. One promising extension is the fully actuated drone system because it is equivalent to controlling a six-DoF floating rigid body system in $SE(3)$. Therefore, the proposed method will be validated in a fully actuated drone system in the future.

Appendix A
Proofs of Lemma

As the proof of the Lemma 1 can be found in [32], we will omit it here. The remaining Lemmas 2, 3, and 4 are proven in this section. To begin with, the derivatives of coefficients α, β, γ, and $\|\xi\|$ are calculated as follows:

$$\frac{d\|\xi\|}{dt} = \frac{\xi^T\dot{\xi}}{\|\xi\|} \qquad \text{for } \xi \in \mathbb{R}^3 \tag{77}$$

$$\dot{\alpha} = \left(1 - \alpha - \frac{\beta}{2}\|\xi\|^2\right)\frac{\xi^T\dot{\xi}}{\|\xi\|^2} \qquad \text{for } \alpha = sc = \frac{\sin\|\xi\|}{\|\xi\|} \tag{78}$$

$$\dot{\beta} = 2(\alpha - \beta)\frac{\xi^T\dot{\xi}}{\|\xi\|^2} \qquad \text{for } \beta = s^2 = \frac{2(1-\cos\|\xi\|)}{\|\xi\|^2} \tag{79}$$

$$\dot{\gamma} = \left(\gamma - \frac{1}{\beta}\right)\frac{\xi^T\dot{\xi}}{\|\xi\|^2} \qquad \text{for } \gamma = \frac{\alpha}{\beta} = \frac{\|\xi\|}{2}\cot\frac{\|\xi\|}{2}. \tag{80}$$

Now, we obtain the relations of the odd and even power terms, i.e., $\mathrm{ad}_\lambda^{2n-1}$ and ad_λ^{2n}. The following claims are summarized with proof.

Claim 1 (Relation of odd power terms $\mathrm{ad}_\lambda^{2n-1}$): For $n \geq 2$

$$\mathrm{ad}_\lambda^{2n-1} = \left[\begin{array}{c|c} \lceil\xi\rceil^{2n-1} & 0_3 \\ \hline 0_3 & \lceil\xi\rceil^{2n-1}\end{array}\right] + \left[\begin{array}{c|c} 0_3 & (*) \\ \hline 0_3 & 0_3\end{array}\right] \tag{81}$$

where

$$\begin{aligned}(*) &= (-1)^{n-1}\|\xi\|^{2(n-1)}\lceil\eta\rceil \\ &\quad {}^{\sim\sim} + (-1)^{n-1}2(n-1)\eta^T\xi\|\xi\|^{2(n-2)}\lceil\xi\rceil.\end{aligned}$$

Proof: Let us prove the claim through mathematical induction.

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(Base case): $n = 2$

$$\begin{aligned}
\mathrm{ad}_\lambda^3 &= \mathrm{ad}_\lambda^2 \times \mathrm{ad}_\lambda \\
&= \left[\begin{array}{c|c} \lceil\xi\rceil^2 & \lceil\xi\rceil\lceil\eta\rceil + \lceil\eta\rceil\lceil\xi\rceil \\ \hline 0_3 & \lceil\xi\rceil^2 \end{array}\right]\left[\begin{array}{c|c} \lceil\xi\rceil & \lceil\eta\rceil \\ \hline 0_3 & \lceil\xi\rceil \end{array}\right] \\
&= \left[\begin{array}{c|c} \lceil\xi\rceil^3 & \lceil\xi\rceil^2\lceil\eta\rceil + (\lceil\xi\rceil\lceil\eta\rceil + \lceil\eta\rceil\lceil\xi\rceil)\lceil\xi\rceil \\ \hline 0_3 & \lceil\xi\rceil^3 \end{array}\right] \\
&= \left[\begin{array}{c|c} \lceil\xi\rceil^3 & (\xi\xi^T - \|\xi\|^2 I_3)\lceil\eta\rceil \\ & \quad + (\xi\eta^T + \eta\xi^T - 2\eta^T\xi I_3)\lceil\xi\rceil \\ \hline 0_3 & \lceil\xi\rceil^3 \end{array}\right] \\
&= \left[\begin{array}{c|c} \lceil\xi\rceil^3 & -\|\xi\|^2\lceil\eta\rceil - 2\eta^T\xi\lceil\xi\rceil \\ \hline 0_3 & \lceil\xi\rceil^3 \end{array}\right]
\end{aligned}$$

in which the following relations are used:

$$\begin{aligned}
&\lceil\xi\rceil^2 = \xi\xi^T - \|\xi\|^2 I_3 \\
&\lceil\xi\rceil\lceil\eta\rceil = \eta\xi^T - \xi^T\eta I_3 \\
&\xi\xi^T\lceil\eta\rceil + \xi\eta^T\lceil\xi\rceil + \eta\xi^T\lceil\xi\rceil = 0_3.
\end{aligned}$$

The last relation is derived as

$$\eta\xi^T\lceil\xi\rceil = 0_3 \ (\because \lceil\xi\rceil\xi = 0_{3\times 1} \Leftrightarrow -\xi^T\lceil\xi\rceil = 0_{1\times 3})$$

$$\xi(\xi^T\lceil\eta\rceil + \eta^T\lceil\xi\rceil) = 0_3 \ (\because \lceil\eta\rceil\xi = -\lceil\xi\rceil\eta).$$

(Induction hypothesis): Assume that the equation is valid for some positive integer $n = k$

$$\mathrm{ad}_\lambda^{2k-1} = \left[\begin{array}{c|c} \lceil\xi\rceil^{2k-1} & 0_3 \\ \hline 0_3 & \lceil\xi\rceil^{2k-1} \end{array}\right] + \left[\begin{array}{c|c} 0_3 & (*) \\ \hline 0_3 & 0_3 \end{array}\right]$$

where

$$\begin{aligned}
(*) &= (-1)^{k-1}\|\xi\|^{2(k-1)}\lceil\eta\rceil \\
&\quad + (-1)^{k-1}2(k-1)\eta^T\xi\|\xi\|^{2(k-2)}\lceil\xi\rceil.
\end{aligned}$$

(Induction step): Check the equation when $n = k + 1$

$$\begin{aligned}
\mathrm{ad}_\lambda^{2k+1} &= \mathrm{ad}_\lambda^{2k-1} \times \mathrm{ad}_\lambda^2 \\
&= \left[\begin{array}{c|c} \lceil\xi\rceil^{2k-1} & (*) \\ \hline 0_3 & \lceil\xi\rceil^{2k-1} \end{array}\right]\left[\begin{array}{c|c} \lceil\xi\rceil^2 & \lceil\xi\rceil\lceil\eta\rceil + \lceil\eta\rceil\lceil\xi\rceil \\ \hline 0_3 & \lceil\xi\rceil^2 \end{array}\right] \\
&= \left[\begin{array}{c|c} \lceil\xi\rceil^{2k+1} & (**) \\ \hline 0_3 & \lceil\xi\rceil^{2k+1} \end{array}\right]
\end{aligned}$$

where

$$\begin{aligned}
(**) &= \lceil\xi\rceil^{2k-1}(\lceil\xi\rceil\lceil\eta\rceil + \lceil\eta\rceil\lceil\xi\rceil) \\
&\quad + (-1)^{k-1}\|\xi\|^{2(k-1)}\lceil\eta\rceil\lceil\xi\rceil^2 \\
&\quad + (-1)^{k-1}2(k-1)\eta^T\xi\|\xi\|^{2(k-2)}\lceil\xi\rceil^3 \\
&= (-1)^{k-1}\|\xi\|^{2(k-1)}(\lceil\xi\rceil^2\lceil\eta\rceil + \lceil\xi\rceil\lceil\eta\rceil\lceil\xi\rceil + \lceil\eta\rceil\lceil\xi\rceil^2) \\
&\quad + (-1)^k 2(k-1)\eta^T\xi\|\xi\|^{2(k-1)}\lceil\xi\rceil \\
&= (-1)^{k-1}\|\xi\|^{2(k-1)}(-\|\xi\|^2\lceil\eta\rceil - 2\eta^T\xi\lceil\xi\rceil) \\
&\quad + (-1)^k 2(k-1)\eta^T\xi\|\xi\|^{2(k-1)}\lceil\xi\rceil \\
&= (-1)^k\|\xi\|^{2k}\lceil\eta\rceil + (-1)^k(2k)\eta^T\xi\|\xi\|^{2(k-1)}\lceil\xi\rceil.
\end{aligned}$$

Therefore, the claim is proven. □

Claim 2 (Relation of even power terms ad_λ^{2n}): For $n \geq 2$

$$\begin{aligned}
\mathrm{ad}_\lambda^{2n} &= (-1)^{n-1}\|\xi\|^{2(n-1)}\mathrm{ad}_\lambda^2 \\
&\quad + (-1)^{n-1}2(n-1)\eta^T\xi\left[\begin{array}{c|c} 0_3 & \|\xi\|^{2(n-2)}\lceil\xi\rceil^2 \\ \hline 0_3 & 0_3 \end{array}\right].
\end{aligned} \tag{82}$$

Proof: Let us prove the claim through mathematical induction.

(Base case): $n = 2$

$$\begin{aligned}
\mathrm{ad}_\lambda^4 &= \mathrm{ad}_\lambda^3 \times \mathrm{ad}_\lambda \\
&= \left[\begin{array}{c|c} \lceil\xi\rceil^3 & -\|\xi\|^2\lceil\eta\rceil - 2\eta^T\xi\lceil\xi\rceil \\ \hline 0_3 & \lceil\xi\rceil^3 \end{array}\right]\left[\begin{array}{c|c} \lceil\xi\rceil & \lceil\eta\rceil \\ \hline 0_3 & \lceil\xi\rceil \end{array}\right] \\
&= \left[\begin{array}{c|c} \lceil\xi\rceil^4 & \lceil\xi\rceil^3\lceil\eta\rceil + \lceil\eta\rceil\lceil\xi\rceil^3 + \lceil\xi\rceil^2\lceil\eta\rceil\lceil\xi\rceil + \lceil\xi\rceil\lceil\eta\rceil\lceil\xi\rceil^2 \\ \hline 0_3 & \lceil\xi\rceil^4 \end{array}\right] \\
&= \left[\begin{array}{c|c} -\|\xi\|^2\lceil\xi\rceil^2 & -\|\xi\|^2(\lceil\xi\rceil\lceil\eta\rceil + \lceil\eta\rceil\lceil\xi\rceil) - 2\eta^T\xi\lceil\xi\rceil^2 \\ \hline 0_3 & -\|\xi\|^2\lceil\xi\rceil^2 \end{array}\right] \\
&= -\|\xi\|^2\mathrm{ad}_\lambda^2 - 2\eta^T\xi\left[\begin{array}{c|c} 0_3 & \lceil\xi\rceil^2 \\ \hline 0_3 & 0_3 \end{array}\right]
\end{aligned}$$

in which the following relations are used

$$\begin{aligned}
&\lceil\xi\rceil^3\lceil\eta\rceil + \lceil\eta\rceil\lceil\xi\rceil^3 = -\|\xi\|^2(\lceil\xi\rceil\lceil\eta\rceil + \lceil\eta\rceil\lceil\xi\rceil) \\
&\lceil\xi\rceil^2\lceil\eta\rceil\lceil\xi\rceil + \lceil\xi\rceil\lceil\eta\rceil\lceil\xi\rceil^2 = -2\eta^T\xi\lceil\xi\rceil^2.
\end{aligned}$$

(Induction hypothesis): Assume that the equation is valid for some positive integer $n = k$.

$$\begin{aligned}
\mathrm{ad}_\lambda^{2k} &= (-1)^{k-1}\|\xi\|^{2(k-1)}\mathrm{ad}_\lambda^2 \\
&\quad + (-1)^{k-1}2(k-1)\eta^T\xi\left[\begin{array}{c|c} 0_3 & \|\xi\|^{2(k-2)}\lceil\xi\rceil^2 \\ \hline 0_3 & 0_3 \end{array}\right].
\end{aligned}$$

(Induction step): Check the equation when $n = k + 1$

$$\begin{aligned}
\mathrm{ad}_\lambda^{2(k+1)} &= \mathrm{ad}_\lambda^{2k} \times \mathrm{ad}_\lambda^2 \\
&= (-1)^{k-1}\|\xi\|^{2(k-1)}\mathrm{ad}_\lambda^4 \\
&\quad + (-1)^{k-1}2(k-1)\eta^T\xi\left[\begin{array}{c|c} 0_3 & \|\xi\|^{2(k-2)}\lceil\xi\rceil^4 \\ \hline 0_3 & 0_3 \end{array}\right] \\
&= (-1)^k\|\xi\|^{2k}\mathrm{ad}_\lambda^2 \\
&\quad + (-1)^k\|\xi\|^{2(k-1)}2\eta^T\xi\left[\begin{array}{c|c} 0_3 & \lceil\xi\rceil^2 \\ \hline 0_3 & 0_3 \end{array}\right] \\
&\quad + (-1)^{k-1}2(k-1)\eta^T\xi\left[\begin{array}{c|c} 0_3 & -\|\xi\|^{2(k-1)}\lceil\xi\rceil^2 \\ \hline 0_3 & 0_3 \end{array}\right] \\
&= (-1)^k\|\xi\|^{2k}\mathrm{ad}_\lambda^2 \\
&\quad + (-1)^k(2k)\eta^T\xi\left[\begin{array}{c|c} 0_3 & -\|\xi\|^{2(k-1)}\lceil\xi\rceil^2 \\ \hline 0_3 & 0_3 \end{array}\right].
\end{aligned}$$

Therefore, the claim is proven. □

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Based on the Claims 1 and 2, the four Lemmas are proven in the following equation.

1) Proof of Lemma 2

Let us start from (15), the formal definition of dexp_λ

$$\mathrm{dexp}_\lambda = \sum_{j=0}^{\infty} \frac{1}{(j+1)!}\mathrm{ad}_\lambda^j = I_6 + \frac{1}{2!}\mathrm{ad}_\lambda + \frac{1}{3!}\mathrm{ad}_\lambda^2 + \cdots$$

where the two diagonal terms are surely identical as

$$I_3 + \frac{1}{2!}\lceil\xi\rceil + \frac{1}{3!}\lceil\xi\rceil^2 + \cdots = \sum_{j=0}^{\infty}\frac{1}{(j+1)!}\lceil\xi\rceil^j = \mathrm{dexp}_\xi$$

and the nonzero off-diagonal term becomes

$$\begin{aligned}&\frac{1}{2!}(\lceil\eta\rceil) + \frac{1}{3!}(\lceil\eta,\xi\rceil) + \frac{1}{4!}(-\|\xi\|^2\lceil\eta\rceil - 2\eta^T\xi\lceil\xi\rceil)\\ &\quad + \frac{1}{5!}(-\|\xi\|^2\lceil\eta,\xi\rceil - 2\eta^T\xi\lceil\xi\rceil^2)\\ &\quad + \frac{1}{6!}(\|\xi\|^4\lceil\eta\rceil + 4\eta^T\xi\|\xi\|^2\lceil\xi\rceil) + \cdots.\end{aligned}$$

Above off-diagonal term is composed of four coefficients $\lceil\eta\rceil$, $\lceil\eta,\xi\rceil$, $\eta^T\xi\lceil\xi\rceil$, and $\eta^T\xi\lceil\xi\rceil^2$. Let us check each coefficient, one by one, the term related to $\lceil\eta\rceil$

$$\begin{aligned}\Rightarrow\ &\frac{1}{2!}\lceil\eta\rceil - \frac{1}{4!}\|\xi\|^2\lceil\eta\rceil + \frac{1}{6!}\|\xi\|^4\lceil\eta\rceil - \cdots\\ &= \left(\frac{1}{2!} - \frac{\|\xi\|^2}{4!} + \frac{\|\xi\|^4}{6!} - \cdots\right)\lceil\eta\rceil\\ &= \frac{1-\cos(\|\xi\|)}{\|\xi\|^2}\lceil\eta\rceil\\ &= \frac{\beta}{2}\lceil\eta\rceil\end{aligned}$$

the term related to $\lceil\eta,\xi\rceil$

$$\begin{aligned}\Rightarrow\ &\frac{1}{3!}\lceil\eta,\xi\rceil - \frac{1}{5!}\|\xi\|^2\lceil\eta,\xi\rceil + \frac{1}{7!}\|\xi\|^4\lceil\eta,\xi\rceil\\ &= \frac{1}{\|\xi\|^3}\left(\frac{\|\xi\|^3}{3!} - \frac{\|\xi\|^5}{5!} + \frac{\|\xi\|^7}{7!} - \cdots\right)\lceil\eta,\xi\rceil\\ &= \frac{1}{\|\xi\|^3}\left(\|\xi\| - \sin(\|\xi\|)\right)\lceil\eta,\xi\rceil\\ &= \frac{1-\alpha}{\|\xi\|^2}\lceil\eta,\xi\rceil\end{aligned}$$

the term related to $\eta^T\xi\lceil\xi\rceil$

$$\begin{aligned}\Rightarrow\ &\left(-\frac{2}{4!} + \frac{4\|\xi\|^2}{6!} - \frac{6\|\xi\|^4}{8!} + \cdots\right)\eta^T\xi\lceil\xi\rceil\\ &= \frac{1}{\|\xi\|^4}\left(-\frac{2\|\xi\|^4}{4!} + \frac{4\|\xi\|^6}{6!} - \frac{6\|\xi\|^8}{8!} + \cdots\right)\eta^T\xi\lceil\xi\rceil\\ &= \frac{1}{\|\xi\|^4}\left((-2+2) + \left(-\frac{2}{2!}+1\right)\|\xi\|^2 + \left(\frac{2}{4!} - \frac{4}{4!}\right)\|\xi\|^4\right.\\ &\quad\left. + \left(-\frac{2}{6!} + \frac{6}{6!}\right)\|\xi\|^6 + \left(\frac{2}{8!} - \frac{8}{8!}\right)\|\xi\|^8 + \cdots\right)\eta^T\xi\lceil\xi\rceil\\ &= \frac{1}{\|\xi\|^4}\left(-2 + 2\left(1 - \frac{\|\xi\|^2}{2!} + \frac{\|\xi\|^4}{4!} - \cdots\right)\right.\\ &\quad\left. + \|\xi\|\left(\|\xi\| - \frac{\|\xi\|^3}{3!} + \frac{\|\xi\|^5}{5!} - \cdots\right)\right)\eta^T\xi\lceil\xi\rceil\\ &= \frac{1}{\|\xi\|^4}(-2 + 2\cos(\|\xi\|) + \|\xi\|\sin(\|\xi\|))\eta^T\xi\lceil\xi\rceil\\ &= \frac{1}{\|\xi\|^2}\left(\frac{\sin(\|\xi\|)}{\|\xi\|} - \frac{2(1-\cos(\|\xi\|))}{\|\xi\|^2}\right)\eta^T\xi\lceil\xi\rceil\\ &= \frac{1}{\|\xi\|^2}(\alpha-\beta)\eta^T\xi\lceil\xi\rceil\end{aligned}$$

the term related to $\eta^T\xi\lceil\xi\rceil^2$

$$\begin{aligned}\Rightarrow\ &-\left(\frac{2}{5!} - \frac{4\|\xi\|^2}{7!} + \frac{6\|\xi\|^2}{9!}\right)\eta^T\xi\lceil\xi\rceil^2\\ &= -\frac{1}{\|\xi\|^5}\left(\frac{2\|\xi\|^5}{5!} - \frac{4\|\xi\|^7}{7!} + \frac{6\|\xi\|^9}{9!}\right)\eta^T\xi\lceil\xi\rceil^2\\ &= -\frac{1}{\|\xi\|^5}\left((-3+1+2)\|\xi\| + \left(\frac{3}{3!} - \frac{3}{3!}\right)\|\xi\|^3\right.\\ &\quad\left. + \left(-\frac{3}{5!} + \frac{5}{5!}\right)\|\xi\|^5 + \left(\frac{3}{7!} - \frac{7}{7!}\right)\|\xi\|^7 + \cdots\right)\eta^T\xi\lceil\xi\rceil^2\\ &= -\frac{1}{\|\xi\|^5}\left(-3\left(\|\xi\| - \frac{\|\xi\|^3}{3!} + \frac{\|\xi\|^5}{5!} - \cdots\right)\right.\\ &\quad\left. + \|\xi\|\left(1 - \frac{\|\xi\|^2}{2!} + \frac{\|\xi\|^4}{4!} - \cdots\right) + 2\|\xi\|\right)\eta^T\xi\lceil\xi\rceil^2\\ &= -\frac{1}{\|\xi\|^5}\left(3(\sin(\|\xi\|)) + \|\xi\|\cos(\|\xi\|) + 2\|\xi\|\right)\eta^T\xi\lceil\xi\rceil^2\\ &= -\frac{1}{\|\xi\|^5}\left(3(\|\xi\| - \sin(\|\xi\|)) - \|\xi\|(1-\cos(\|\xi\|))\right)\eta^T\xi\lceil\xi\rceil^2\\ &= -\frac{1}{\|\xi\|^2}\left(\frac{3(1-\alpha)}{\|\xi\|^2} - \frac{\beta}{2}\right)\eta^T\xi\lceil\xi\rceil^2\end{aligned}$$

and finally, the off-diagonal term denoted by $C_\xi(\eta)$ is rearranged as

$$\begin{aligned}\therefore\ C_\xi(\eta) \triangleq\ &\frac{\beta}{2}\lceil\eta\rceil + \frac{1-\alpha}{\|\eta\|^2}(\lceil\eta\rceil\lceil\xi\rceil + \lceil\xi\rceil\lceil\eta\rceil)\\ &+ \frac{\alpha-\beta}{\|\eta\|^2}\xi^T\eta\lceil\xi\rceil - \left(\frac{3(1-\alpha)}{\|\xi\|^4} - \frac{\beta}{2\|\xi\|^2}\right)\xi^T\eta\lceil\xi\rceil^2.\end{aligned}$$

Furthermore, let us begin with (16), the definition of $\mathrm{dexp}_\lambda^{-1}$

$$\begin{aligned}\mathrm{dexp}_\lambda^{-1} &= \sum_{j=0}^{\infty}\frac{B_j}{j!}\mathrm{ad}_\lambda^j\\ &= I_6 + B_1\mathrm{ad}_\lambda + \sum_{k=1}^{\infty}\frac{B_{2k}}{(2k)!}\mathrm{ad}_\lambda^{2k}\\ &= I_6 + B_1\mathrm{ad}_\lambda\end{aligned}$$

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$$+\frac{1}{\|\xi\|^2}\underbrace{\left(\sum_{k=1}^{\infty}\frac{B_{2k}}{(2k)!}(-1)^{k-1}\|\xi\|^{2k}\right)}_{\text{(i)}}\mathrm{ad}_\lambda^2$$

$$+\left[\begin{array}{c|c} 0_3 & \underbrace{\sum_{k=1}^{\infty}\frac{B_{2k}}{(2k)!}(-1)^{k-1}2(k-1)\eta^T\xi\|\xi\|^{2(k-2)}\lceil\xi\rceil^2}_{\text{(ii)}} \\ \hline 0_3 & 0_3 \end{array}\right].$$

Note that the Claim 2 is used in the derivation, and the Bernoulli number vanishes for every odd $n>1$, i.e., $B_n=0$ for $n=3,5,7,\ldots$. The terms (i) and (ii) are calculated as

$$\begin{aligned}
\text{(i)} &= \sum_{k=1}^{\infty}\frac{B_{2k}}{(2k)!}(-1)^{k-1}\|\xi\|^{2k} = 1-\gamma \\
\text{(ii)} &= \sum_{k=1}^{\infty}\frac{B_{2k}}{(2k)!}(-1)^{k-1}2(k-1)\eta^T\xi\|\xi\|^{2(k-2)} \\
&= \frac{\eta^T\xi}{\|\xi\|^3}\sum_{k=1}^{\infty}\frac{B_{2k}}{(2k)!}(-1)^{k-1}(2k)\|\xi\|^{2k-1} \\
&\quad -\frac{2\eta^T\xi}{\|\xi\|^4}\sum_{k=1}^{\infty}\frac{B_{2k}}{(2k)!}(-1)^{k-1}\|\xi\|^{2k} \\
&= \frac{\eta^T\xi}{\|\xi\|^3}\left(\frac{\partial(1-\gamma)}{\partial\|\xi\|}\right)-\frac{2\eta^T\xi}{\|\xi\|^4}(1-\gamma) \\
&= \frac{\eta^T\xi}{\|\xi\|^3}\left(-\frac{\gamma}{\|\xi\|}+\frac{1}{\|\xi\|\beta}\right)-\frac{2\eta^T\xi}{\|\xi\|^4}(1-\gamma) \\
&= \left(-\gamma+\frac{1}{\beta}-2(1-\gamma)\right)\frac{\eta^T\xi}{\|\xi\|^4} \\
&= \frac{\gamma+(1/\beta)-2}{\|\xi\|^4}\eta^T\xi
\end{aligned}$$

where the followings are used:

$$\begin{aligned}
\gamma(y) &= \frac{\alpha}{\beta} = \frac{c(y)}{s(y)} = \frac{(y/2)\cos(y/2)}{\sin(y/2)} \\
\frac{\partial\gamma}{\partial y} &= \frac{1}{2}\frac{\cos(y/2)}{\sin(y/2)} \\
&\quad +\frac{y}{2}\frac{-\sin(y/2)\frac{1}{2}\sin(y/2)-\cos(y/2)\frac{1}{2}\cos(y/2)}{\sin^2(y/2)} \\
&= \frac{1}{2}\frac{\cos(y/2)}{\sin(y/2)}-\frac{y}{2}\frac{1/2}{\sin^2(y/2)} \quad (\sin^2+\cos^2=1) \\
&= \frac{\gamma(y)}{y}-\frac{1}{y\beta(y)}.
\end{aligned}$$

Two diagonal terms are identical as

$$I_3-\frac{1}{2}\lceil\xi\rceil+\frac{1-\gamma}{\|\xi\|^2}\lceil\xi\rceil^2 = \mathrm{dexp}_\xi^{-1}.$$

The nonzero off-diagonal term denoted by $D_\xi(\eta)$ is rearranged into

$$\begin{aligned}
\therefore\ D_\xi(\eta) \triangleq &-\frac{1}{2}\lceil\eta\rceil+\frac{1-\gamma}{\|\xi\|^2}(\lceil\xi\rceil\lceil\eta\rceil+\lceil\eta\rceil\lceil\xi\rceil) \\
&+\frac{\gamma+(1/\beta)-2}{\|\xi\|^4}\eta^T\xi\lceil\xi\rceil^2.
\end{aligned}$$

This is the end of the proof of Lemma 2.

2) Proof of Lemma 3

Since $\mathrm{dexp}_\xi = I_3+\frac{1}{2}\beta\lceil\xi\rceil+\frac{1-\alpha}{\|\xi\|^2}\lceil\xi\rceil^2$, time derivative can be calculated by chain rule. Consider the derivative term by term

$$\begin{aligned}
\frac{d}{dt}\mathrm{dexp}_\xi = &\underbrace{\frac{1}{2}\dot\beta\lceil\xi\rceil}_{\text{(i)}}+\frac{1}{2}\beta\lceil\dot\xi\rceil \\
&+\underbrace{\frac{d}{dt}\left(\frac{1-\alpha}{\|\xi\|^2}\right)\lceil\xi\rceil^2}_{\text{(ii)}}+\frac{1-\alpha}{\|\xi\|^2}(\lceil\dot\xi\rceil\lceil\xi\rceil+\lceil\xi\rceil\lceil\dot\xi\rceil)
\end{aligned}$$

where

$$\begin{aligned}
\text{(i)} &= \frac{1}{2}\dot\beta\lceil\xi\rceil = \frac{1}{2}\cdot 2(\alpha-\beta)\frac{\xi^T\dot\xi}{\|\xi\|^2}\lceil\xi\rceil = \frac{\alpha-\beta}{\|\xi\|^2}\xi^T\dot\xi\lceil\xi\rceil \\
\text{(ii)} &= \frac{d}{dt}\left(\frac{1-\alpha}{\|\xi\|^2}\right)\lceil\xi\rceil^2 \\
&= \frac{1}{\|\xi\|^4}\left(-\dot\alpha\|\xi\|^2-2(1-\alpha)\|\xi\|\frac{\xi^T\dot\xi}{\|\xi\|}\right)\lceil\xi\rceil^2 \\
&= \frac{1}{\|\xi\|^4}\left(-\left(1-\alpha-\frac{\beta}{2}\|\xi\|^2\right)\frac{\xi^T\dot\xi}{\|\xi\|^2}\|\xi\|^2\right. \\
&\qquad \left.-2(1-\alpha)\xi^T\dot\xi\right)\lceil\xi\rceil^2 \\
&= \frac{1}{\|\xi\|^4}\left(-3(1-\alpha)+\frac{\beta}{2}\|\xi\|^2\right)\xi^T\dot\xi\lceil\xi\rceil^2 \\
&= -\left(\frac{3(1-\alpha)}{\|\xi\|^4}-\frac{\beta}{2\|\xi\|^2}\right)\xi^T\dot\xi\lceil\xi\rceil^2.
\end{aligned}$$

The time derivative of differential of exponential map on $SO(3)$ becomes equal to $C_\xi(\dot\xi)$

$$\begin{aligned}
\therefore\ \frac{d}{dt}\mathrm{dexp}_\xi = &\frac{\beta}{2}\lceil\dot\xi\rceil+\frac{1-\alpha}{\|\xi\|^2}(\lceil\dot\xi\rceil\lceil\xi\rceil+\lceil\xi\rceil\lceil\dot\xi\rceil) \\
&+\frac{\alpha-\beta}{\|\xi\|^2}\xi^T\dot\xi\lceil\xi\rceil \\
&-\left(\frac{3(1-\alpha)}{\|\xi\|^4}-\frac{\beta}{2\|\xi\|^2}\right)\xi^T\dot\xi\lceil\xi\rceil^2 \\
=: &\ C_\xi(\dot\xi).
\end{aligned}$$

Also, since ${\mathrm{dexp}_\xi}^{-1} = I_3-\frac{1}{2}\lceil\xi\rceil+\frac{1-\gamma}{\|\xi\|^2}\lceil\xi\rceil^2$

$$\frac{d}{dt}{\mathrm{dexp}_\xi}^{-1} = -\frac{1}{2}\lceil\dot\xi\rceil+\frac{(1-\gamma)}{\|\xi\|^2}(\lceil\dot\xi\rceil\lceil\xi\rceil+\lceil\xi\rceil\lceil\dot\xi\rceil)$$

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$$+\underbrace{\frac{d}{dt}\left(\frac{1-\gamma}{\|\xi\|^2}\right)\lceil\xi\rceil^2}_{\text{(iii)}}$$

where

$$\begin{aligned}
\text{(iii)} &= \frac{d}{dt}\left(\frac{1-\gamma}{\|\xi\|^2}\right)\lceil\xi\rceil^2 \\
&= \frac{-\dot{\gamma}\|\xi\|^2-(1-\gamma)\cdot 2\|\xi\|\frac{\xi^T\dot{\xi}}{\|\xi\|}}{\|\xi\|^4}\lceil\xi\rceil^2 \\
&= \frac{1}{\|\xi\|^4}\left(-\left(\gamma-\frac{1}{\beta}\right)\frac{\xi^T\dot{\xi}}{\|\xi\|^2}\|\xi\|^2-2(1-\gamma)\xi^T\dot{\xi}\right)\lceil\xi\rceil^2 \\
&= \frac{1}{\|\xi\|^2}\frac{(\frac{1}{\beta}+\gamma-2)}{\|\xi\|^2}\xi^T\dot{\xi}\lceil\xi\rceil^2.
\end{aligned}$$

The time derivative of inverse differential of exponential map on $SO(3)$ becomes equal to $D_\xi(\dot{\xi})$

$$\begin{aligned}
\therefore\ \frac{d}{dt}\mathrm{dexp}_\xi{}^{-1} &= -\frac{1}{2}\lceil\dot{\xi}\rceil+\frac{1-\gamma}{\|\xi\|^2}(\lceil\dot{\xi}\rceil\lceil\xi\rceil+\lceil\xi\rceil\lceil\dot{\xi}\rceil) \\
&\quad+\frac{1}{\|\xi\|^2}\frac{\frac{1}{\beta}+\gamma-2}{\|\xi\|^2}\xi^T\dot{\xi}\lceil\xi\rceil^2 \\
&=: D_\xi(\dot{\xi}).
\end{aligned}$$

This is the end of the proof of Lemma 3.

3) Proof of Lemma 4

For future use, let us differentiate the given Γ_i functions with respect to time: for Γ_1

$$\begin{aligned}
\dot{\Gamma}_1 &= \frac{d}{dt}\left(\frac{1-\alpha}{\|\xi\|^2}\right) \\
&= \frac{1}{\|\xi\|^4}\left(-\dot{\alpha}\|\xi\|^2-(1-\alpha)\cdot 2\|\xi\|\frac{\xi^T\dot{\xi}}{\|\xi\|}\right) \\
&= \frac{1}{\|\xi\|^4}\left(-\left(1-\alpha-\frac{\beta}{2}\|\xi\|^2\right)-2(1-\alpha)\right)\xi^T\dot{\xi} \\
&= \Gamma_3\xi^T\dot{\xi} = \frac{1}{\|\xi\|^2}\left(\frac{\beta}{2}-3\Gamma_1\right)\xi^T\dot{\xi}
\end{aligned}$$

for Γ_2

$$\begin{aligned}
\dot{\Gamma}_2 &= \frac{d}{dt}\left(\frac{\alpha-\beta}{\|\xi\|^2}\right) \\
&= \frac{1}{\|\xi\|^4}\left((\dot{\alpha}-\dot{\beta})\|\xi\|^2-(\alpha-\beta)\cdot 2\|\xi\|\frac{\xi^T\dot{\xi}}{\|\xi\|}\right) \\
&= \frac{1}{\|\xi\|^4}\left(\left(1-\alpha-\frac{\beta}{2}\|\xi\|^2-2(\alpha-\beta)\right)-2(\alpha-\beta)\right)\xi^T\dot{\xi} \\
&= \frac{1}{\|\xi\|^4}\left((1-\alpha)-4(\alpha-\beta)-\frac{\beta}{2}\|\xi\|^2\right)\xi^T\dot{\xi} \\
&= \left(\Gamma_1-4\Gamma_2-\frac{\beta}{2}\right)\frac{\xi^T\dot{\xi}}{\|\xi\|^2}
\end{aligned}$$

for Γ_3

$$\begin{aligned}
\dot{\Gamma}_3 &= \frac{d}{dt}\left(\frac{\frac{\beta}{2}-3\Gamma_1(\xi)}{\|\xi\|^2}\right) \\
&= \frac{1}{\|\xi\|^4}\left(\left(\frac{\dot{\beta}}{2}-3\dot{\Gamma}_1\right)\|\xi\|^2-\left(\frac{\beta}{2}-3\Gamma_1\right)2\|\xi\|\frac{\xi^T\dot{\xi}}{\|\xi\|}\right) \\
&= \frac{1}{\|\xi\|^4}\left((\alpha-\beta)-3\left(\frac{\beta}{2}-3\Gamma_1\right)-(\beta-6\Gamma_1)\right)\xi^T\dot{\xi} \\
&= \frac{1}{\|\xi\|^2}\left(\frac{\alpha-\beta}{\|\xi\|^2}+\frac{5(3\Gamma_1-\frac{\beta}{2})}{\|\xi\|^2}\right)\xi^T\dot{\xi} \\
&= (\Gamma_2-5\Gamma_3)\frac{\xi^T\dot{\xi}}{\|\xi\|^2}
\end{aligned}$$

for Γ_4

$$\dot{\Gamma}_4 = \frac{d}{dt}\left(\frac{1-\gamma}{\|\xi\|^2}\right) = \Gamma_5\xi^T\dot{\xi}$$

for Γ_5

$$\begin{aligned}
\dot{\Gamma}_5 &= \frac{d}{dt}\left(\frac{1}{\|\xi\|^2}\frac{\frac{1}{\beta}+\gamma-2}{\|\xi\|^2}\right) \\
&= \frac{1}{\|\xi\|^8}\left(\left(-\frac{\dot{\beta}}{\beta^2}+\dot{\gamma}\right)\|\xi\|^4-\left(\frac{1}{\beta}+\gamma-2\right)4\|\xi\|^3\frac{\xi^T\dot{\xi}}{\|\xi\|}\right) \\
&= \frac{1}{\|\xi\|^6}\left(\left(-\frac{2(\alpha-\beta)}{\beta^2}+\gamma-\frac{1}{\beta}\right)-4\left(\frac{1}{\beta}+\gamma-2\right)\right)\xi^T\dot{\xi} \\
&= \frac{1}{\|\xi\|^6}\left(2\left(1-\frac{\gamma}{\beta}\right)-3\left(\frac{1}{\beta}+\gamma-2\right)\right)\xi^T\dot{\xi} \\
&= \left(\frac{2(1-\frac{\gamma}{\beta})}{\|\xi\|^4}-\frac{3(\frac{1}{\beta}+\gamma-2)}{\|\xi\|^4}\right)\frac{\xi^T\dot{\xi}}{\|\xi\|^2} \\
&= \left(\frac{2(1-\frac{\gamma}{\beta})}{\|\xi\|^4}-3\Gamma_5\right)\frac{\xi^T\dot{\xi}}{\|\xi\|^2}.
\end{aligned}$$

Since $C_\xi(\eta)$ can be expressed in terms of Γ_i functions as follows:

$$C_\xi(\eta) = \frac{\beta}{2}\lceil\eta\rceil+\Gamma_1(\xi)\lceil\eta,\xi\rceil+\Gamma_2(\xi)\xi^T\eta\lceil\xi\rceil+\Gamma_3(\xi)\xi^T\eta\lceil\xi\rceil^2$$

its time-derivative is described by

$$\begin{aligned}
\frac{d}{dt}C_\xi(\eta) &= \frac{\beta}{2}\lceil\dot{\eta}\rceil+\underbrace{\Gamma_1(\xi)\frac{d}{dt}(\lceil\eta,\xi\rceil)}_{\text{(i)}}+\underbrace{\Gamma_2(\xi)\frac{d}{dt}(\xi^T\eta\lceil\xi\rceil)}_{\text{(ii)}} \\
&\quad+\underbrace{\Gamma_3(\xi)\frac{d}{dt}(\xi^T\eta\lceil\xi\rceil^2)}_{\text{(iii)}}+\underbrace{\frac{\dot{\beta}}{2}\lceil\eta\rceil}_{\text{(iv)}}+\underbrace{\dot{\Gamma}_1(\xi)\lceil\eta,\xi\rceil}_{\text{(v)}} \\
&\quad+\underbrace{\dot{\Gamma}_2(\xi)\xi^T\eta\lceil\xi\rceil}_{\text{(vi)}}+\underbrace{\dot{\Gamma}_3(\xi)\xi^T\eta\lceil\xi\rceil^2}_{\text{(vii)}}
\end{aligned}$$

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where

$$\begin{aligned}
\text{(i)} &= \Gamma_1 \frac{d}{dt}(\lceil\eta\rceil\lceil\xi\rceil + \lceil\xi\rceil\lceil\eta\rceil) \\
&= \Gamma_1(\lceil\dot{\eta},\xi\rceil + \lceil\eta,\dot{\xi}\rceil) \\
\text{(ii)} &= \Gamma_2((\dot{\xi}^T\eta + \xi^T\dot{\eta})\lceil\xi\rceil + \xi^T\eta\lceil\dot{\xi}\rceil) \\
\text{(iii)} &= \Gamma_3((\dot{\xi}^T\eta + \xi^T\dot{\eta})\lceil\xi\rceil^2 + \xi^T\eta\lceil\xi,\dot{\xi}\rceil) \\
\text{(iv)} &= \frac{\dot{\beta}}{2}\lceil\eta\rceil = \Gamma_2\xi^T\dot{\xi}\lceil\eta\rceil \\
\text{(v)} &= \dot{\Gamma}_1\lceil\eta,\xi\rceil = \Gamma_3\xi^T\dot{\xi}\lceil\eta,\xi\rceil \\
\text{(vi)} &= \dot{\Gamma}_2\xi^T\eta\lceil\xi\rceil = \left(\Gamma_1 - 4\Gamma_2 - \frac{\beta}{2}\right)\zeta(\eta)\lceil\xi\rceil \\
\text{(vii)} &= \dot{\Gamma}_3\xi^T\eta\lceil\xi\rceil^2 = (\Gamma_2 - 5\Gamma_3)\zeta(\eta)\lceil\xi\rceil^2.
\end{aligned}$$

To sum up all terms

$$\begin{aligned}
\Rightarrow\ & \frac{\beta}{2}\lceil\dot{\eta}\rceil - \frac{\beta}{2}\zeta(\eta)\lceil\xi\rceil + \Gamma_1(\lceil\dot{\eta},\xi\rceil + \lceil\eta,\dot{\xi}\rceil + \zeta(\eta)\lceil\xi\rceil) \\
& + \Gamma_2(\xi^T\eta\lceil\dot{\xi}\rceil + \xi^T\dot{\xi}\lceil\eta\rceil \\
& + (\dot{\xi}^T\eta + \xi^T\dot{\eta} - 4\zeta(\eta))\lceil\xi\rceil + \zeta(\eta)\lceil\xi\rceil^2) \\
& + \Gamma_3(\xi^T\eta\lceil\xi,\dot{\xi}\rceil + \xi^T\dot{\xi}\lceil\eta,\xi\rceil + (\dot{\xi}^T\eta + \xi^T\dot{\eta} - 5\zeta(\eta))\lceil\xi\rceil^2)
\end{aligned}$$

finally, we have the time-derivative of $C_\xi(\eta)$

$$\begin{aligned}
\therefore\ \frac{d}{dt}C_\xi(\eta) &= \frac{\beta}{2}(\lceil\dot{\eta}\rceil - \zeta(\eta)\lceil\xi\rceil) \\
&\quad + \Gamma_1(\lceil\dot{\eta},\xi\rceil + \lceil\eta,\dot{\xi}\rceil + \zeta(\eta)\lceil\xi\rceil) \\
&\quad + \Gamma_2(\delta_1(\eta,\dot{\eta}) + \zeta(\eta)\lceil\xi\rceil^2) \\
&\quad + \Gamma_3\delta_2(\eta,\dot{\eta}) \\
&=: \dot{C}_{\xi,\dot{\xi}}(\eta,\dot{\eta}).
\end{aligned}$$

Also, since $D_\xi(\eta) = -\frac{1}{2}\lceil\eta\rceil + \Gamma_4(\xi)\lceil\eta,\xi\rceil + \Gamma_5(\xi)\xi^T\eta\lceil\xi\rceil^2$, its time derivative is obtained by

$$\begin{aligned}
\frac{d}{dt}D_\xi(\eta) = & -\frac{1}{2}\lceil\dot{\eta}\rceil + \underbrace{\Gamma_4(\xi)\frac{d}{dt}(\lceil\eta,\xi\rceil)}_{\text{(i)}} \\
& + \underbrace{\Gamma_5(\xi)\frac{d}{dt}(\xi^T\eta\lceil\xi\rceil^2)}_{\text{(ii)}} + \underbrace{\dot{\Gamma}_4(\xi)\lceil\eta,\xi\rceil}_{\text{(iii)}} \\
& + \underbrace{\dot{\Gamma}_5(\xi)\xi^T\eta\lceil\xi\rceil^2}_{\text{(iv)}}
\end{aligned}$$

where

$$\begin{aligned}
\text{(i)} &= \Gamma_4(\lceil\dot{\eta},\xi\rceil + \lceil\eta,\dot{\xi}\rceil) \\
\text{(ii)} &= \Gamma_5((\dot{\xi}^T\eta + \xi^T\dot{\eta})\lceil\xi\rceil^2 + \xi^T\eta\lceil\xi,\dot{\xi}\rceil) \\
\text{(iii)} &= \dot{\Gamma}_4\lceil\eta,\xi\rceil = \Gamma_5\xi^T\dot{\xi}\lceil\eta,\xi\rceil \\
\text{(iv)} &= \dot{\Gamma}_5\xi^T\eta\lceil\xi\rceil^2 = \left(\frac{2}{\|\xi\|^2}\frac{1-\frac{\gamma}{\beta}}{\|\xi\|^2} - 3\Gamma_5\right)\xi^T\eta\frac{\xi^T\dot{\xi}}{\|\xi\|^2}\lceil\xi\rceil^2 \\
&= \left(\frac{2}{\|\xi\|^2}\frac{1-\frac{\gamma}{\beta}}{\|\xi\|^2} - 3\Gamma_5\right)\zeta(\eta)\lceil\xi\rceil^2.
\end{aligned}$$

To sum up all terms

$$\begin{aligned}
\Rightarrow\ & -\frac{1}{2}\lceil\dot{\eta}\rceil + \frac{2}{\|\xi\|^2}\frac{1-\frac{\gamma}{\beta}}{\|\xi\|^2}\zeta(\eta)\lceil\xi\rceil^2 \\
& + \Gamma_4(\lceil\dot{\eta},\xi\rceil + \lceil\eta,\dot{\xi}\rceil) \\
& + \Gamma_5(\xi^T\eta\lceil\xi,\dot{\xi}\rceil + \xi^T\dot{\xi}\lceil\eta,\xi\rceil + (\dot{\xi}^T\eta + \xi^T\dot{\eta} - 3\zeta(\eta))\lceil\xi\rceil^2)
\end{aligned}$$

finally, we have the time-derivative of $D_\xi(\eta)$

$$\begin{aligned}
\therefore\ \frac{d}{dt}D_\xi(\eta) = & -\frac{1}{2}\lceil\dot{\eta}\rceil + \frac{2}{\|\xi\|^2}\frac{(1-\frac{\gamma}{\beta})}{\|\xi\|^2}\zeta(\eta)\lceil\xi\rceil^2 \\
& + \Gamma_4(\lceil\dot{\eta},\xi\rceil + \lceil\eta,\dot{\xi}\rceil) + \Gamma_5\delta_3(\eta,\dot{\eta}) \\
=: & \ \dot{D}_{\xi,\dot{\xi}}(\eta,\dot{\eta}).
\end{aligned}$$

This is the end of the proof of Lemma 4.

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